High-Weight Cosets of Generalized and Extended Reed–Solomon Codes

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Abstract

Let a redundancy-r generalized Reed–Solomon code have evaluation set $S = \mathbf{P}^1(\mathbb{F}_q) \setminus A$, where $|A| = s$, and arbitrary nonzero column multipliers. For $r \geq 6$, $\operatorname{char} \mathbb{F}_q > r - 1$, and

$$q \geq 6(r+s) - 16 + \left\lfloor 2\sqrt{6(r+s)-18} \right\rfloor,$$

we classify every coset of weight at least $r-1$. If $s = 0$, the covering radius is $r-1$, and its maximum-weight directions are the tangent and conjugate-secant points of the normal rational curve. If $s > 0$, the radius is r: omitted curve points have weight r, while tangents, conjugate secants, and rational secants incident with A give exactly weight $r-1$. We count both shells and classify their MDS and NMDS one-column extensions and family-aggregate weight enumerators.

Associate to a syndrome f its Hankel kernel W_f; a split squarefree degree-$(r-2)$ member supported on S is exactly an $(r-2)$-column representation. Simultaneous choice of $r-5$ factors contracts to one cubic pencil. A genus-one count supplies the remaining factors outside explicit rank-two and small-characteristic exceptional loci. Thus, in every characteristic, possible high-weight syndromes lie in those loci, while witnesses are abundant elsewhere. Exact low-redundancy refinements and public artifacts delimit small-field and trust boundaries.

Index Terms

coset weight, covering radius, deep holes, generalized Reed–Solomon codes, Hankel matrices, MDS codes, near-MDS codes, normal rational curves.

I. Introduction

The weight of a coset of a linear code is the minimum Hamming weight of a vector in that coset. We determine the cosets of weight at least $r-1$ for a family of long generalized and extended Reed–Solomon codes of redundancy r. The same classification determines their deep holes and the one-column extensions whose Singleton defect is at most one.

Let $A \subseteq \mathbb{P}^1(\mathbb{F}_q)$, put $s = |A|$ and $S = \mathbb{P}^1(\mathbb{F}_q) \setminus A$, and let $C_S(v)$ be a redundancy-r generalized Reed–Solomon code on S, with arbitrary nonzero coordinate multipliers v. A parity-check matrix H_S has columns proportional to the points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

For a syndrome f, let $d_S(f)$ be the least number of columns of H_S whose span contains f; this is the weight of the corresponding coset. Column multipliers do not change these projective spans.

Theorem I.1 (High-weight cosets)**.** Let $r \geq 6$, let $A \subseteq \mathbb{P}^1(\mathbb{F}_q)$ have size s, and assume $r \leq q + 1 - s$. Put

$$Q^*_{r,s} = 6(r+s) - 16 + \left\lfloor 2\sqrt{6(r+s)-18} \right\rfloor.$$

If $q \geq Q^*_{r,s}$, then

$$f \notin \mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p} \quad \Longrightarrow \quad d_S(f) \leq r-2, \qquad p = \operatorname{char} \mathbb{F}_q, \tag{1}$$

where $\mathcal{P}_r$ is the rank-at-most-two catalecticant locus and

$$\mathcal{M}^{\max}_{r,p} = \mathbb{P}\left\langle e_j : 0 \leq j \leq r-1, \quad \binom{r-2}{j} = \binom{r-2}{j-1} = 0 \pmod p \right\rangle,$$

where $\binom{n}{k} = 0$ for $k \notin \{0, \ldots, n\}$. In characteristic two the same conclusion holds with $6(r+s) - 22 + \lfloor 2\sqrt{6(r+s)-24} \rfloor$ in place of $Q^*_{r,s}$.

Assume in addition that $p > r-1$. If $s = 0$, the covering radius is $r-1$, and the cosets of weight $r-1$ are exactly the tangent and conjugate-secant directions. If $s > 0$, the covering radius is r: the weight-r directions are the omitted curve points, and the weight-$r-1$ directions are exactly the tangent points, conjugate-secant points, and interior points of rational secants incident with A. Their projective counts are

$$N_r = s, \qquad N_{r-1} = \frac{q(q+1)^2}{2} + \frac{(q-1)s(2q+1-s)}{2}.$$

The corresponding numbers of nonzero cosets are $(q-1)N_r$ and $(q-1)N_{r-1}$.

Here $Q^*_{r,s} = 6(r+s)+O(\sqrt{r+s})$. The conclusion has three levels: in every characteristic the field-size hypothesis confines all weight-at-least-$(r-1)$ syndromes to $\mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}$; if $p > r-1$, the Lucas carrier vanishes and the shells are completely classified; in small characteristic, the only unresolved arbitrary-redundancy arithmetic is confined to the explicit Lucas carrier $\mathcal{M}^{\max}_{r,p}$. In the complete-classification range, deleting even one point is substantive: the radius jumps from $r-1$ to r, and the omitted points become exactly the new weight-r shell.

The only computer-algebra input to the uniform theorem is the exact reduced terminal-carrier decomposition in Proposition IV.3, verified by the registered Gröbner-elimination certificate. Recursive component selection, simultaneous finite-field selection, shell transfer, and the enumerative double counts are mathematical arguments. The covering-radius and MDS-extension gates are imported from Seroussi–Roth and Dür.

Coding interpretation. Theorem I.1 classifies every coset in the two highest possible weight shells of these GRS codes and, equivalently, every one-column extension having Singleton defect at most one. Indeed, appending f as a parity-check column gives a code $\widehat{C}_f$ with $d(\widehat{C}_f) = d_S(f)+1$. A family-wise double count determines the aggregate number of minimum words in the tangent, conjugate-secant, and incident-split-secant families; the standard NMDS recurrence then gives their complete family-aggregate weight enumerators.

For $s = 0$, $C_S(v)$ is monomially equivalent to the full extended or projective Reed–Solomon code. For $s > 0$, a projective change of coordinate moves an omitted point to infinity, so the code is monomially equivalent to an ordinary GRS code. To our knowledge, no earlier result classifies, for arbitrary redundancy, every projective syndrome direction of such a code with coset weight at least $r-1$ after finitely many prescribed points are deleted. In the range of Theorem I.1, these are exactly the omitted curve points in weight r, and the tangent, conjugate-secant, and deleted-point-incident split-secant directions in weight $r-1$.

The qualification is essential. The covering-radius, syndrome, and MDS-extension dictionaries are classical [1], [2], [3], [4]. For full projective support at redundancy four, the tangent and conjugate-secant shell and its count are already known [5], [6], [7]. Point-deleted projective evaluation sets were studied in [8], and recent work gives general MDS/NMDS constructions from deep holes [9]. None of these results gives the complete point-deleted top two shells at arbitrary redundancy. The novelty here is that conjunction and its uniform exclusion off the explicit carrier; the extension and enumerator statements are its consequences.

The witness sought is a degree-$(r-2)$ binary form in W_f whose distinct rational roots all lie in S; by the Hankel criterion, its roots are the $r-2$ parity-check columns spanning f. Instead of classifying every intermediate redundancy, the proof selects $r-5$ of these linear factors simultaneously. Contraction leaves a redundancy-five (R5) pencil of binary cubics, and any split squarefree member avoiding the selected roots and A completes the desired form. The carrier theorem confines the cases in which every such contraction is trapped to the persistent rank-two locus $\mathcal{P}_r$ or the small-characteristic Lucas locus $\mathcal{M}^{\max}_{r,p}$; outside that union, a degree-six selector, a Vandermonde choice, and the exact R5 count produce the cubic. The witnesses are abundant: there are at least $q^{r-4}/(r-2)! - O_{r,s}(q^{r-9/2})$ split locators supported on S.

The full-support calculations remain useful where they are sharper than the uniform theorem. Redundancy five supplies the terminal cubic engine and a complete classification over every prime power $q \geq 7$; redundancies six and seven supply exact small-field and modular refinements. They are stated and proved after the arbitrary-redundancy theorem, with their separate radius and certificate boundaries kept explicit. The earlier R8–R10 calculations are companion records, not induction hypotheses or claims of this submission.

II. Results at a Glance

Theorem I.1 classifies every coset of weight at least $r-1$. Its conclusion is independent of the placement of the deleted set A and of all nonzero GRS column multipliers.

TABLE I
THE HIGH-WEIGHT COSETS WHEN $p > r-1$ AND THE FIELD-SIZE HYPOTHESIS OF THEOREM I.1 HOLDS. COUNTS ARE PROJECTIVE SYNDROME DIRECTIONS; LITERAL NONZERO COSET COUNTS ARE LARGER BY A FACTOR OF $q-1$.

Support	Radius	Weight r	Weight $r-1$
$s=0$	$r-1$	none	tangents and conjugate secants; count $q(q+1)^2/2$
$s>0$	r	omitted curve points; count s	tangents, conjugate secants, and interiors of split secants incident with A; count $q(q+1)^2/2 + (q-1)s(2q+1-s)/2$

Appending a syndrome direction as a parity-check column raises its coset weight by one. The preceding table therefore gives all one-column extensions with Singleton defect at most one:

$d_S(f)$	$d(\ker[H_S \mid f])$	extension class
r	$r+1$	MDS
$r-1$	r	NMDS
$\leq r-2$	$\leq r-1$	defect at least two.

Theorem VI.2 then counts the minimum supports in each NMDS family and determines its complete aggregate weight enumerator.

In arbitrary characteristic the exact Lucas-carrier arithmetic can be more complicated, but the general conclusion remains short:

$$d_S(f) \geq r-1 \quad \Longrightarrow \quad f \in \mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}.$$

The carrier is explicit from adjacent zero entries in Pascal row $r-2$. The fixed R5–R7 classifications decide its arithmetic in the ranges stated in Section VIII; they sharpen the uniform theorem in small fields but are not inputs to it.

A. Why the classification closes

The proof has three steps. The Hankel kernel W_f records the degree-$(r-2)$ binary forms whose roots can represent f: a squarefree product of rational linear factors supported on S is exactly an $(r-2)$-column representation. Choosing one factor contracts both the syndrome and the witness problem by one degree; choosing $r-5$ factors leaves a redundancy-five pencil of binary cubics. The recursive carrier theorem identifies when every such contraction is trapped. Everywhere else, one degree-six selector chooses the factors simultaneously, and the exact R5 pencil count supplies a split squarefree cubic avoiding those factors and A.

$$\begin{gathered}
f \notin \underbrace{\mathcal{P}_r}_{\text{persistent rank-two locus}} \cup \underbrace{\mathcal{M}^{\max}_{r,p}}_{\text{small-characteristic Lucas locus}} \\
\Downarrow \quad \text{simultaneous selection and contraction} \\
\text{one split squarefree cubic in the terminal Hankel pencil,} \\
\text{avoiding the markers and } A \\
\Downarrow \quad \text{lift through } r-5 \text{ distinct markers in } S \\
W_f \text{ contains a split squarefree form supported on } S \\
\Updownarrow \quad \text{Hankel criterion} \\
d_S(f) \leq r-2.
\end{gathered}$$

Fig. 1. The proof route. The carrier theorem identifies the only possible obstruction; simultaneous contraction constructs a locator everywhere else.

The bounded-field rows below use one finite-certificate contract. The certificate enumerates projective syndromes modulo the stated group, records canonical representatives, stabilizers, orbit sizes, and split-member data relative to the stated evaluation set, and checks that the orbit sizes exhaust the ambient projective space. A separate replay recomputes those invariants from the records. These finite steps complete the split-free geometric classification only in the named fields; covering-radius promotion to deep holes remains a separate cited input.

A first-pass reader can now continue with the coding dictionary in Section III, the terminal R5 count in Section VII, the recursive carrier in Section IV, and the simultaneous escape theorem in Section V. The detailed fixed-level polar packages, exceptional orbit tables, modular calculations, and verification records are collected in the appendices.

III. The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. If x, y is a basis of E with dual basis X, Y, our perfect divided-power/symmetric-power pairing is

$$\left\langle x^{[n-i]}y^{[i]}, X^{n-j}Y^j \right\rangle = \delta_{ij}, \qquad \left\langle \sum_{i=0}^{n} a_i x^{[n-i]}y^{[i]}, \sum_{i=0}^{n} b_i X^{n-i}Y^i \right\rangle = \sum_{i=0}^{n} a_i b_i.$$

Thus the pairing is coefficient extraction, with no binomial factors in any characteristic. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

Read $f = (a_0, \ldots, a_{r-1})$ as a finite sequence. A form $g \in W_f$ is the characteristic polynomial of a recurrence satisfied by that sequence, while a squarefree split g has a basis of geometric-sequence solutions. Thus split-free means that the syndrome sequence satisfies no order-$(r-2)$ recurrence with squarefree completely split characteristic polynomial.

Lemma III.1 (Hankel criterion)**.** Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\big) \quad \Longleftrightarrow \quad g \in W_f.$$

TABLE II
NOTATION USED FROM THE HANKEL DICTIONARY ONWARD.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in the full-support code $\mathrm{PRS}(k)$
r	redundancy; $r = \lvert S\rvert - \dim C_S(v)$, and $r = q+1-k$ only for full support
$n = r-1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \mathrm{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, the geometric sequences $\nu(\lambda_i)$ solve the recurrence with characteristic polynomial g; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary III.2 (Split-free criterion)**.** Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We use *split-free* without a covering-radius hypothesis and call a direction *shallow* when W_f contains a completely split squarefree form.

The following terms describe the interfaces used in the recursive proof. They name geometric data, not additional hypotheses.

TABLE III
THE RECURSIVE VOCABULARY AND ITS CODING ROLE.

Term	Meaning
split locator on S	a product of $r-2$ distinct rational linear factors in W_f, with zeros in S; it encodes an $(r-2)$-column representation of f
split-free / shallow	no split locator on S / the existence of one; for full support this is exactly the absence / existence of a completely split squarefree member of W_f
marker	one selected factor λ; contraction replaces f by $\iota_\lambda f$, and lifting multiplies by λ
polar line	the projective image of $[\lambda] \mapsto [\iota_\lambda f]$ as the marker varies
collision	a marker fixed in the lower witness system, so every lift repeats λ
persistent locus	common-quadratic rank-two families; tangents and conjugate secants are split-free, while rational secants are shallow on full support
carrier	a closed locus where every contraction may remain obstructed; membership does not by itself assert deepness

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy. If H_f has rank one, then $\dim W_f = r-2$, which is incompatible with a common quadratic factor because the space of degree-$(r-2)$ forms divisible by that factor has dimension $r-3$. Otherwise $\dim W_f = r-3$, and

$$r-3 \le r-1-\deg\gcd(W_f),$$

so its degree is at most two.

A. Coding consequences and recognition

A split squarefree degree-$(r-2)$ member of W_f, with roots in the evaluation support, is an explicit coset representative of weight at most $r-2$. Its absence is a deep-hole condition only after the independent covering-radius gate. Section VI applies this dictionary to deleted supports and arbitrary GRS multipliers, classifies the weight-r and weight-$(r-1)$ shells, and translates them into MDS and NMDS one-column extensions. In the full-support case the radius is $r-1$, so the MDS extension class is empty; omitted curve points supply that class as soon as the support is proper.

The same dictionary gives recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects the modular locus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table IV and recorded in the supplement.

Recognition at the fixed redundancies proved here begins with row-reduction of H_f and the listed family-membership tests. Coherent polar induction replaces a direct search through the projective members of W_f.

IV. The recursive carrier theorem

Composite contraction reduces the uniform problem to the geometry of linear spaces contained in the terminal bad locus. We determine that locus after reduction in each geometric fiber and show that all modular descendants merge into one linear carrier.

For $r \geq 6$, put $n = r-1$, and let k be the ground field of characteristic $p > 0$. In divided-power coordinates define

$$\mathcal{M}_{r,p}^{\max} = \mathbb{P}\left\langle e_j : 0 \leq j \leq r-1, \quad \binom{r-2}{j} = \binom{r-2}{j-1} = 0 \pmod{p} \right\rangle, \tag{2}$$

where $\binom{n}{k} = 0$ when $k \notin \{0, \dots, n\}$. An empty span denotes the empty projective scheme. Let

$$C_f^{(2)} : \operatorname{Sym}^{r-3} E^\vee \longrightarrow \Gamma^2 E, \qquad R \longmapsto \iota_R f,$$

be the second catalecticant. For $f = (a_0 : \cdots : a_{r-1})$, its matrix is

$$\begin{pmatrix} a_0 & a_1 & \cdots & a_{r-3} \\ a_1 & a_2 & \cdots & a_{r-2} \\ a_2 & a_3 & \cdots & a_{r-1} \end{pmatrix}.$$

Define $\mathcal{P}_r = \{[f] \in \mathbb{P}(\Gamma^{r-1} E) : \operatorname{rank} C_f^{(2)} \leq 2\}$. This is the second secant variety, with its determinantal scheme structure. Its rank-two points split into tangent, rational-secant, and conjugate-secant types; only the first and third are split-free on full support.

Lemma IV.1 (Base-point freeness off the persistent carrier)**.** For every $r \geq 5$, if $[f] \notin \mathcal{P}_r$, then W_f has trivial geometric gcd. If W_f has nontrivial gcd, its degree is one exactly when H_f has rank one, on the normal rational curve, and is two when H_f has rank two.

Proof. Work over an algebraic closure and suppose that a linear form L divides every member of W_f. Multiplication by L is injective, and, with

$$K_L = (LE^\vee)^\perp \subset \Gamma^2 E,$$

the perfect pairing gives

$$\{R \in \operatorname{Sym}^{r-3} E^\vee : LR \in W_f\} = \left(C_f^{(2)}\right)^{-1}(K_L).$$

Here K_L is one-dimensional. Because $W_f \subset L \operatorname{Sym}^{r-3} E^\vee$, multiplication by L identifies the left-hand side with W_f, so its dimension is at least $r-3$. If $C_f^{(2)}$ had rank three, it would be surjective and the right-hand side would instead have dimension $(r-2-3)+1 = r-4$. Hence $\operatorname{rank} C_f^{(2)} \leq 2$, so $[f] \in \mathcal{P}_r$, a contradiction.

For the final assertion, the rank-one normal form is a curve point and gives $W_f = L \operatorname{Sym}^{r-3} E^\vee$. At a rank-two secant or tangent point, direct substitution in the Hankel matrix gives respectively $W_f = L_1 L_2 \operatorname{Sym}^{r-4} E^\vee$ or $W_f = L^2 \operatorname{Sym}^{r-4} E^\vee$. □

The next two propositions and one theorem prove the unconditional geometric layer: they find every reduced terminal component, merge all modular descendants, and select the components that persist up the contraction tower. On a first pass, read their statements and the four-row component table in Theorem IV.7, then resume at Theorem IV.9. The intervening elimination and characteristic-wise substitutions prove exhaustiveness and add no hypothesis to the finite-depth theorem.

Definition IV.2 (Reduced terminal bad scheme)**.** Let $L = \mathbb{P}\langle F, G\rangle \subset \mathbb{P}(\operatorname{Sym}^3 E^\vee)$ be a pencil of binary cubics, and let $z = (z_0 : \cdots : z_5) \in \operatorname{Gr}(2,4) \subset \mathbb{P}^5$ be the Plücker point given by the 2×2 minors of the coefficient matrix of F, G. On the affine root chart, for distinct roots t, s, the determinant $F(t)G(s) - F(s)G(t)$ vanishes exactly when one member of L contains both roots. After dividing out its diagonal factor, the affine equation is

$$\Phi_z(t,s) = z_0 + z_1(t+s) + z_2(t^2+ts+s^2) + z_3 ts + z_4 ts(t+s) + z_5 t^2 s^2.$$

We use its bihomogenization on $\mathbb{P}^1 \times \mathbb{P}^1$, so the cases $t = \infty$ and/or $s = \infty$ are included. Let $K \subset \mathbb{Z}[z_0, \dots, z_5]$ be the ideal

$$\begin{aligned}(3z_4^2 - 9z_2z_5 - z_3z_5, z_3z_4 - 3z_1z_5, z_3^2 - 9z_0z_5,\\ z_2z_3 - z_1z_4 + z_0z_5, z_1z_3 - 3z_0z_4, 3z_1^2 - 9z_0z_2 - z_0z_3).\end{aligned}$$

The ideal K is the saturated coefficient-elimination ideal for the symmetric residual factorization of this off-diagonal incidence; the anti-invariant collision component is removed before elimination. Thus $V_{\mathbb{P}^5}(K) \cap \operatorname{Gr}(2,4)$ records the residual factorization used in the non-S_3 monodromy argument below. Certificate SC checks the coefficient-elimination identity defining K; the monodromy interpretation is proved in the text.

Now let $c = (c_0 : \cdots : c_4) \in \mathbb{P}(\Gamma^4 E)$ be a terminal syndrome. On the open set where H_c has rank two, its Hankel kernel W_c is a cubic pencil, and its Plücker point is

$$z(c) = (c_2c_4 - c_3^2, -c_1c_4 + c_2c_3, c_1c_3 - c_2^2, c_0c_4 - c_1c_3, -c_0c_3 + c_1c_2, c_0c_2 - c_1^2).$$

When H_c has rank one, every coordinate of $z(c)$ vanishes; this base locus lies in $\mathcal{P}_5$ and is included separately below. Here z^*K denotes the homogeneous ideal obtained by substituting these six quadratic expressions into the generators of K, not a pullback across the rank-one base locus. The determinant

$$D = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix},$$

cuts out the persistent rank-at-most-two locus $\mathcal{P}_5$. Saturating by D removes that component. In characteristic p, define

$$I_{\mathrm{res},p} = \sqrt{\left((z^*K) \otimes \mathbb{F}_p : D^\infty\right)}, \qquad \mathcal{B}_{5,p}^{\mathrm{red}} = V(D) \cup V(I_{\mathrm{res},p}).$$

This is the *reduced terminal bad scheme*. Thus the scheme being decomposed is fixed before its component calculation; it is not inferred from a finite-field census.

Proposition IV.3 (Computer-assisted terminal decomposition)**.** The reduced terminal bad scheme of Definition IV.2 is the irredundant union of the prime cubic $V(D)$ and one residual prime. Away from characteristics two and three the residual prime is the projected Veronese surface

$$[u : v : w] \longmapsto [6u^2 : 3uv : v^2 + 2uw : 3vw : 6w^2], \tag{3}$$

the locus of syndrome quartics that are scalar multiples of squares of binary quadratics. The coordinates in (3) are the bottom c_i; writing $z_i = \binom{4}{i} c_i$ for the plain coefficients of the associated quartic $f_c = \sum_{i=0}^4 z_i X^{4-i} Y^i$, the parametrization gives $f_c = 6(uX^2 + vXY + wY^2)^2$, so the same surface is the plain-coefficient Veronese in rescaled divided-power coordinates. In characteristic two it is the plane $V(c_0, c_4)$. In characteristic three it is the cone with c_2 free and prime ideal

$$(c_1c_3 - c_0c_4,\ c_1^3 - c_0^2c_3,\ c_0c_3^2 - c_1^2c_4,\ c_3^3 - c_1c_4^2). \tag{4}$$

There is no flat reduced integral model having exactly these residual fibers.

Every terminal point outside this scheme has a base-point-free separable pencil: base-point freeness follows from Lemma IV.1, while the only inseparable Hankel pencil is the characteristic-three all-cube nucleus, contained in the residual prime. For such a pencil, lying outside the scheme is equivalent to geometric S_3 monodromy of its cubic map. Thus Proposition VII.6 and Lemma VII.7 apply to every terminal point outside $\mathcal{B}_{5,p}^{\mathrm{red}}$.

Proof. The elimination has a concrete geometric source. On $\mathrm{Gr}(2, \Gamma^3 E)$, use the Plücker coordinates and ordered-pair equation of Definition IV.2. For the Hankel matrix

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix},$$

the signed maximal minors give the map explicitly:

$$z(c) = (c_2c_4 - c_3^2,\ -c_1c_4 + c_2c_3,\ c_1c_3 - c_2^2, c_0c_4 - c_1c_3,\ -c_0c_3 + c_1c_2,\ c_0c_2 - c_1^2).$$

After the diagonal and collision factors already charged as deletion divisors are saturated away, the terminal bad locus is the pullback, along the Hankel Plücker map $c \mapsto z(c)$, of the factorization locus of Φ_z. The involution $(t, s) \leftrightarrow (s, t)$ leaves three possibilities. Symmetric factors give the residual component below; exchanged factors give $D = 0$; and an anti-invariant factor is proportional to $s - t$. In the last case the only nonzero Plücker coordinates are $z_2 = -\mu$ and $z_3 = 3\mu$, and hence

$$z_0z_5 - z_1z_4 + z_2z_3 = (-\mu)(3\mu) = -3\mu^2.$$

Thus it survives only in characteristic three and is then the diagonal collision already removed. Thus the two displayed primes are the only possible noncollision components before small-characteristic specialization.

The exact saturated bridge can also be read without the elimination file. Let $J = z^*K$ in the notation of Definition IV.2. If V denotes the seven-generator residual ideal printed below, exact reduction gives $J \subseteq V$, $3D\,V \subseteq J$, and $(J : D^\infty) = V$ over $\mathbb{Q}$. The factor 3 is necessary over $\mathbb{Z}$, which is precisely why the characteristic-three fiber is checked separately.

Outside characteristics two and three, elimination of the ordered residual roots gives the prime kernel of (3); one convenient generating set is

$$
\begin{aligned}
&2c_3^3 - 3c_2c_3c_4 + c_1c_4^2,\\
&6c_2c_3^2 - 9c_2^2c_4 + 2c_1c_3c_4 + c_0c_4^2,\\
&2c_1c_3^2 - 3c_1c_2c_4 + c_0c_3c_4, \quad c_0c_3^2 - c_1^2c_4,
\end{aligned}
$$

together with their images under $c_i \leftrightarrow c_{4-i}$. Direct reduction gives the plane in characteristic two and (4) in characteristic three. The ordered-Hessian calculation sends its complementary ruling to rank one.

Here exhaustiveness is an exact elimination statement, not a visual factorization heuristic. Certificate SC starts from the terminal ordered-root incidence ideal, saturates the collision factors, eliminates the root variables, and verifies the two generic primes and their characteristic-$2,3$ fibers by Gröbner reduction. Its public Python and Singular inputs are in `supplement/evidence/stable-components/`; every generator and remainder used for this conclusion is hash-pinned there.

Each displayed residual ideal is prime: (3) is a projective parametrization with prime kernel, the characteristic-two ideal is linear, and (4) is the toric prime of the monomial quartic after projection from e_2. None is contained in $V(D)$: use $(3,0,1,0,3)$, which is the image of $(u,v,w) = (1,0,1)$ and has $D = 8$, together with e_2 and $(1,0,1,0,0)$, respectively. Thus the intersection with (D) is an irredundant reduced decomposition and has no embedded prime. The residual degree is four generically and one in characteristic two. Constancy of the Hilbert polynomial in a flat projective family rules out the final assertion. The exact eliminations and the characteristic-$2,3,5$ reductions are replayed by the stable-component certificates described in Section C.

Lemma IV.1 makes every pencil outside $V(D)$ base-point-free. An inseparable cubic pencil can occur only in characteristic three; after source and target changes it is $\langle X^3, Y^3\rangle$. The overlapping Hankel rows then give the syndrome e_2, whose $\mathrm{GL}(E)$-orbit lies in the characteristic-three residual prime.

Finally, a base-point-free separable cubic map has transitive geometric monodromy C_3 or S_3. Its off-diagonal ordered-pair curve is reducible exactly in the cyclic case, and the noncollision reducibility is precisely the saturated residual factorization encoded by K. A pencil outside the displayed scheme therefore has geometric monodromy S_3, which proves the last clause. □

For every extension Ω/k, let $U_{f,m}(\Omega)$ be the open set of ordered, pairwise distinct markers in $\mathbb{P}(E_\Omega^\vee)^m$ for which every successive contraction is nonzero.

Definition IV.4 (Coherent polar flag)**.** A coherent polar flag from redundancy r to redundancy five is a sequence

$$f = f_r, \qquad f_{j-1} = \iota_{\lambda_j} f_j \quad (j = r, \dots, 6),$$

over an algebraic closure, where the marker tuple $([\lambda_r], \dots, [\lambda_6])$ lies in the ordered marker space $U_{f,r-5}(\bar{k})$. Thus the markers are distinct and every displayed contraction is nonzero. At level f_{j-1}, for any field Ω over which the flag is defined, an Ω-admissible locator is a split squarefree $g \in W_{f_{j-1}}(\Omega)$ with $\lambda_k \nmid g$ for every retained marker $k = j, \dots, r$. This avoidance makes the iterated lift $\lambda_r \cdots \lambda_j g \in W_f(\Omega)$ squarefree by Equation (20). The polar image at level f_j is the image of $[\lambda] \mapsto [\iota_\lambda f_j]$ on its nonzero domain, a point or line parametrizing the possible next contracted syndrome directions. A coherent lift reverses the displayed sequence.

For $0 \le a < d$, the order-a nucleus of the degree-d normal rational curve is

$$\mathcal{N}_{d,a} = \mathbb{P}\left\langle e_i : \binom{\ell}{i} = 0 \pmod p \text{ for every } \ell = a+1, \dots, d \right\rangle \subset \mathbb{P}(\Gamma^d E);$$

this is the standard coordinate description [10]. If $\widetilde{\mathcal{N}}_{d,a}$ is its affine cone, its one-step coherent lift is

$$\mathcal{M}_{d+1}(\mathcal{N}_{d,a}) = \mathbb{P}\ker\Big(\Gamma^{d+1}E \longrightarrow \mathrm{Hom}(E^\vee, \Gamma^d E/\widetilde{\mathcal{N}}_{d,a})\Big).$$

Iterating this operation places a nucleus from any lower level in $\mathbb{P}(\Gamma^{r-1}E)$.

Proposition IV.5 (Maximal Lucas union)**.** Take every $\mathcal{N}_{d,a}$ with $4 \le d \le r-2$, every $a < d$, and every iterated coherent lift to $\mathbb{P}(\Gamma^{r-1}E)$. Each lies in $\mathcal{M}_{r,p}^{\max}$, and their reduced union equals that single linear scheme. A polar image not contained in the union meets it in degree at most one. Hence a one-parameter contraction family cannot be generically trapped in the Lucas carrier unless its entire polar image is contained there. At an endpoint degree $d = p^b + 1$ over $\mathbb{F}_{p^e}$, the associated rational Lucas block family is nonempty exactly when $b \mid e$, and its blocks are the projective $\mathbb{F}_{p^b}$-sublines of $\mathbb{P}^1(\mathbb{F}_{p^e})$ [11, Prop. 11].

Proof. The displayed coordinate description shows that the support of $\mathcal{N}_{d,a}$ is contained in

$$\widehat{S}_{d,p} = \{i : \binom{d}{i} = 0 \pmod p\}, \tag{5}$$

because its defining conditions include the row $\ell = d$. If a linear nucleus has support $I \subset \{0, \ldots, d\}$, its one-step coherent lift has support

$$L_d(I) = \{j : (j = d+1 \text{ or } j \in I),\ (j = 0 \text{ or } j-1 \in I)\}. \tag{6}$$

Every further coherent lift applies the same operator. It preserves inclusions, and Pascal's identity gives $L_d(\widehat{S}_{d,p}) \subset \widehat{S}_{d+1,p}$. Thus the support obtained from any level d is contained in the support obtained at every later level. Conversely, the order-$(r-3)$ nucleus at $d = r-2$ occurs in the union, and its one-step lift is exactly (2). The union is therefore one linear scheme. Pulling its linear ideal back to a projective line gives nonzero linear forms unless the line is contained, proving the degree bound. The final rational endpoint clause is the cited result of Wang–Wu–Hu; the carrier equality, Pascal nesting, and Hankel incidence above are independent of it. $\square$

Definition IV.6 (Reduced recursively contained carrier)**.** Put $m = r-5$. For a geometric upper syndrome $[f] \in \mathbb{P}(\Gamma^{r-1}E_\Omega)$ over any extension Ω/k, define the terminal contraction map

$$\Psi_f : U_{f,m}(\Omega) \longrightarrow \mathbb{P}(\Gamma^4 E_\Omega), \qquad ([\lambda_r], \ldots, [\lambda_6]) \longmapsto [\iota_{\lambda_6} \cdots \iota_{\lambda_r} f].$$

Let $U^\circ_{f,m}$ be the open subset obtained by deleting the pullbacks of the collision divisor characterized in Lemma A.4 at every contraction stage. Let $X \subset \mathbb{P}(\Gamma^{r-1}E_{\bar{k}})$ be an irreducible closed subvariety, and let $\bar{\eta}_X$ be a geometric generic point over an algebraic closure of the function field $\bar{k}(X)$. Require that

$$\overline{\Psi_{\bar{\eta}_X}(U^\circ_{\bar{\eta}_X,m})}$$

lie in one geometric irreducible component of $\mathcal{B}^{\mathrm{red}}_{5,p}$. Call X *terminal-contained* when it is maximal by inclusion among subvarieties satisfying these conditions. The *reduced recursively contained carrier* is their reduced union together with every iterated lift of every intermediate nucleus. Collision divisors remain deletion conditions rather than carrier components.

Theorem IV.7 (Recursive carrier)**.** The reduced recursively contained carrier generated by the terminal carrier and all intermediate nuclei is exactly

$$\mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}. \tag{7}$$

If a retained marker becomes a root of a lower locator, the lifted locator has a repeated factor; this is exactly the collision divisor of Lemma A.4. This condition is handled by deletion bounds, not by (7).

Proof. Put $m = r-5$, form the marker product $H = \lambda_r \cdots \lambda_6 \in \operatorname{Sym}^m E^\vee$, and let h be its coefficient row. The coordinate row of the bottom syndrome f_5 is

$$h\,\mathrm{Cat}_{m,4}(f), \qquad \mathrm{Cat}_{m,4}(f) = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_m & a_{m+1} & a_{m+2} & a_{m+3} & a_{m+4} \end{pmatrix}. \tag{8}$$

Thus composite contraction is the projective linear map

$$\chi_f : \mathbb{P}(\operatorname{Sym}^m E^\vee) \dashrightarrow \mathbb{P}(\Gamma^4 E), \qquad [h] \longmapsto [h\,\mathrm{Cat}_{m,4}(f)].$$

Over an algebraic closure, products of distinct linear forms are dense in the projective space of binary forms. Deleting the projective kernel of the map χ_f does not change its image closure. Away from the whole-line collision alternative of Lemma A.5, the all-stage collision pullbacks are proper divisors, so deleting them also leaves a dense open. Consequently

$$\overline{\Psi_f(U^\circ_{f,m})} = \mathbb{P}\,\mathrm{Row}(\mathrm{Cat}_{m,4}(f)).$$

This row space is irreducible. If it lies in the finite reduced union of Proposition IV.3, its generic point lies in one component, so the whole row space lies in that component. This is a geometric-marker statement; no density of the finite set of rational marker tuples is used.

For the remaining component check, let $R = \mathbb{P}\,\mathrm{Row}(\mathrm{Cat}_{m,4}(f))$. The middle column below lists the projective linear spaces R that can lie in each terminal component; the last column gives the resulting upper-syndrome locus.

terminal component	contained row space R	upper locus
$V(D)$	point or line ($\dim R \le 1$)	$\mathcal{P}_r$
Veronese ($p \ne 2, 3$)	point	rank one
wild cone ($p = 3$)	point or ruling through e_2	rank one
$V(c_0, c_4)$ ($p = 2$)	binary-plane subspace	$\mathcal{M}^{\max}_{r,2}$

To check the rows, write $[d] = (d_0 : \cdots : d_5) \in \mathbb{P}(\Gamma^5 E)$ and $c_i = xd_i + yd_{i+1}$. As $(x : y)$ varies, $c(x, y)$ parametrizes the one-step polar image of d, which is a point or line.

Persistent component. After substituting $c(x,y)$ into D, the coefficients in x,y are exactly the 3×3 minors of

$$\begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ d_1 & d_2 & d_3 & d_4 \\ d_2 & d_3 & d_4 & d_5 \end{pmatrix},$$

which is the consecutive Fano identity. Thus a polar image contained in $V(D)$ has upper syndrome in $\mathcal{P}_6$; iteration gives $\mathcal{P}_r$.

Generic residual component. For $p \neq 2,3$, substituting the line into the residual prime and equating coefficients gives an ideal whose unique minimal prime is the ideal of the 2×2 minors of $\left(\begin{smallmatrix} d_0 & \cdots & d_4 \\ d_1 & \cdots & d_5 \end{smallmatrix}\right)$, so the upper syndrome has rank one. Equivalently, the projected Veronese contains no projective line.

Characteristic-three residual component. For a ruling of the wild cone through e_2, the coefficient ideal is again the same rank-one ideal. Hence every positive-dimensional ruling lifts into $\mathcal{P}_r$ and creates no new upper component.

Characteristic-two residual component. For the terminal plane $V(c_0,c_4) = \mathbb{P}\langle e_1,e_2,e_3\rangle$, the coefficient ideal has exactly the two minimal primes

$$(d_0,d_1,d_4,d_5) \quad \text{and} \quad (d_3d_4+d_2d_5, d_1d_4+d_0d_5, d_3^2+d_1d_5, d_2d_3+d_0d_5, d_2^2+d_0d_4, d_1d_2+d_0d_3).$$

The second defines a sublocus of the persistent 3×3-minor scheme. The first is the one-step descendant $\mathbb{P}\langle e_2,e_3\rangle$. The successive support chain is

$$\mathbb{P}\langle e_1,e_2,e_3\rangle \longmapsto \mathbb{P}\langle e_2,e_3\rangle \longmapsto \mathbb{P}\langle e_3\rangle \longmapsto \varnothing.$$

Equation (6) identifies this chain with the Lucas lifts. Together with Proposition IV.5, it places every surviving modular lift in $\mathcal{M}_{r,2}^{\max}$.

The coherent-Fano part of Certificate SC verifies these consecutive-row substitutions, including the characteristic-three ruling and the characteristic-two successive supports, against the same incidence ideal. This exhausts every component and proves (7) by induction on the flag length. ☐

Proposition IV.8 (Persistent count and orbit law)**.** Put $n = r-1$ and

$$T = \{z \in \mathbb{F}_{q^2}^{\times} : z^{q+1} = 1\}.$$

The split-free part of $\mathcal{P}_r(\mathbb{F}_q)$ consists of the tangent and conjugate-secant directions and has cardinality $q(q+1)^2/2$. On the conjugate-secant part, the $\mathrm{PGL}_2(q)$-orbits are the inversion orbits on T/T^n; semilinear orbits further identify $[z] \sim [z^p]$. On a tangent fiber, the unipotent stabilizer sends the affine coordinate z to $z+nu$, $u \in \mathbb{F}_q$. Hence the tangent part is one $\mathrm{PGL}_2(q)$-orbit when $p \nmid n$. No complete tangent-orbit classification is asserted here when $p \mid n$.

Proof. The tangent/conjugate-secant identification and count agree with the established full-support calculation [3, Prop. 1][5, Thms. I.4–I.6]; we record the same rank-two geometry uniformly in r, together with its elementary stabilizer action. For a fixed irreducible quadratic, the rank-two syndrome fiber has all $q+1$ projective points. For a fixed square, the rank-one endpoint is removed and q points remain. There are $q(q-1)/2$ irreducible quadratics and $q+1$ squares up to scale, so

$$\frac{q(q-1)}{2}(q+1) + (q+1)q = \frac{q(q+1)^2}{2}.$$

After normalizing the two roots, the conjugate-secant stabilizer is the norm-one torus T, acting by n-th powers; its normalizer inverts T, and field Frobenius raises its elements to the p-th power. The tangent normalization gives the displayed additive action. These are the same uniform rank-two calculations instantiated at redundancies six and seven in Section B. ☐

Theorem IV.9 (Arbitrary-redundancy carrier theorem)**.** Let $r \geq 6$, and let q be a prime power with

$$q \geq Q_r^* := 6r - 16 + \left\lfloor 2\sqrt{6r-18} \right\rfloor. \tag{9}$$

Thus $Q_r^* = Q_{r,0}^*$, the full-support specialization of the simultaneous-contraction threshold $Q_{r,s}^*$. For full support, every split-free redundancy-r syndrome lies in $\mathcal{P}_r \cup \mathcal{M}_{r,p}^{\max}$. If $p > r-1$, the modular term is empty, and the projective deep-hole syndromes are exactly the persistent tangent and conjugate-secant families, with cardinality and semilinear orbit law given by Proposition IV.8. Here the deep-hole identification uses the Seroussi–Roth extension theorem [1, Thm. 1] and Dür's covering-radius equivalence [2]. In characteristic two the carrier containment holds under $q \geq 6r - 22 + \lfloor 2\sqrt{6r-24} \rfloor$.

Proof. Theorem IV.7 identifies the complete reduced carrier. Theorem V.4, with $s = 0$, gives a split squarefree kernel member at every point outside it. Its genus-one point estimate guarantees a split cubic fiber after all prescribed deletions; solving the positivity inequality gives (9). The proof does not assume classifications at redundancies $6,\dots,r-1$.

If $p > r-1$, no interior entry of Pascal row $r-2$ vanishes, so $\mathcal{M}_{r,p}^{\max} = \varnothing$. Rank one and rational secants are shallow; the remaining rank-two points are precisely the tangent and conjugate-secant families and are split-free. The threshold lies in the Seroussi–Roth–Dür radius range, so split-free and deep coincide. The count and orbit law are Proposition IV.8, whose uniform calculation is instantiated, not assumed, at redundancies six and seven. ☐

The carrier theorem is fiberwise and reduced. It makes no assertion about nilpotent structure in a chosen integral incidence model, and in small characteristic it classifies the only possible carrier of additional split-free points, not the arithmetic of every point on that carrier. The simultaneous-marker theorem proves escape from its complement directly.

Proposition IV.10 (Upper-level radius gate)**.** For the ranges used below,

$$\begin{aligned}\rho(\mathrm{PRS}(q-7)) &= 7 \quad (q \geq 43),\\ \rho(\mathrm{PRS}(q-8)) &= 8 \quad (q \geq 53),\\ \rho(\mathrm{PRS}(q-9)) &= 9 \quad (q \geq 59).\end{aligned}$$

More generally, the radius is $r-1$ under the full-support, large-characteristic hypotheses of Theorem I.1.

Proof. The dual GRS code has dimension r. In each displayed range, and also under (9), one has

$$r \leq \lfloor q/2 \rfloor + 2.$$

Seroussi–Roth [1, Thm. 1] therefore gives nonextendability of the corresponding normal rational curve; its stated even-field dimension-three exception is irrelevant because $r \geq 8$. Dür's completeness–covering-radius equivalence [2], in the form recorded by Kaipa [3, Sec. IV and Prop. 4(1)], gives the asserted radius. Thus the promotion from split-free syndromes to deep holes in the companion R8–R10 calculations uses one explicit imported gate. □

V. Simultaneous contraction and escape

The recursive carrier theorem identifies the syndromes whose entire terminal row space is trapped. We now treat every other syndrome without choosing a sequence of intermediate good fibers. All contracted roots are selected at once, and only the terminal R5 pencil is counted.

The contraction identity and finite-field grid lemma below are elementary. What is new here is their coupling to the recursive carrier through one terminal selector: the selector replaces all intermediate-redundancy casework, while the final enumerative input is the redundancy-five split-member count and branch budget of Section VII. Its relation to the twisted-cubic line literature is spelled out in Remark VII.14.

Let E be two-dimensional over a field F. For $0 \leq m \leq r-2$, $f \in \Gamma^{r-1}E$, and $R \in \mathrm{Sym}^m(E^\vee)$, define $\iota_R f \in \Gamma^{r-1-m}E$ by

$$\langle \iota_R f, h\rangle = \langle f, Rh\rangle \qquad \big(h \in \mathrm{Sym}^{r-1-m} E^\vee\big). \tag{10}$$

Proposition V.1 (Composite contraction)**.** For every $h \in \mathrm{Sym}^{r-2-m} E^\vee$,

$$h \in W_{\iota_R f} \quad\Longleftrightarrow\quad Rh \in W_f.$$

The construction commutes with base change and the natural $\mathrm{GL}(E)$-action. If R is split and squarefree, then Rh is split and squarefree exactly when h is split and squarefree and has no root in common with R.

Proof. For every $\lambda \in E^\vee$,

$$\langle \iota_R f, \lambda h\rangle = \langle f, R\lambda h\rangle.$$

These are precisely the two Hankel equations defining the two witness systems. The pairing also proves base-change and equivariance. The final assertion is the usual coprimality criterion after passage to an algebraic closure. □

We use one elementary finite-field lemma to choose the roots of R.

Lemma V.2 (Vandermonde grid)**.** Let $P \in \mathbb{F}_q[x_1, \ldots, x_m]$ be nonzero and satisfy $\deg_{x_i} P \leq d$ for every i. If $q > d+m-1+s$, then there are pairwise distinct $a_1, \ldots, a_m \in \mathbb{F}_q$, outside any prescribed set of s field elements, for which $P(a_1, \ldots, a_m) \neq 0$.

Proof. Multiply P by

$$\prod_{i<j}(x_i - x_j) \prod_{i=1}^{m} \prod_{a \in A_0} (x_i - a),$$

where $A_0 \subset \mathbb{F}_q$ is the forbidden set. The product is nonzero and has degree at most $d+m-1+s < q$ in each variable. A polynomial reduced to degree below q in every variable cannot vanish on all of $\mathbb{F}_q^m$. At a nonzero value, the coordinates are distinct and avoid A_0. □

Put $m = r-5$. If $R = \lambda_1 \cdots \lambda_m$, the coefficient row of R sends f to the bottom syndrome

$$\iota_R f = h\,\mathrm{Cat}_{m,4}(f).$$

Lemma V.3 (Terminal selector)**.** Let E be defined over $\mathbb{F}_q$, let $0 \neq f \in \Gamma^{r-1}E$, and let L_f be the projectivized row space of $\mathrm{Cat}_{m,4}(f)$. If $[f] \notin \mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}$, there is a homogeneous polynomial F_f over $\mathbb{F}_q$, of degree at most six (at most four when $p = 2$), which vanishes on the reduced terminal carrier but whose restriction to L_f is not the zero polynomial. Consequently

$$S_f(\lambda_1, \ldots, \lambda_m) = F_f(\iota_{\lambda_1 \cdots \lambda_m} f)$$

is a nonzero multihomogeneous $\mathbb{F}_q$-polynomial. On an affine marker chart, its dehomogenization has degree at most six, respectively four, in each marker coordinate.

Proof. Theorem IV.7 says that L_f is not contained in the reduced terminal carrier. By Proposition IV.3, that carrier is $V(D) \cup V(I_{\mathrm{res},p})$, where D has degree three and $I_{\mathrm{res},p}$ has generators over the prime field of degree at most three, or degree one when $p = 2$. The irreducible space L_f is contained in neither component. Thus $D|_{L_f} \neq 0$, and some listed generator $G \in I_{\mathrm{res},p}$ has $G|_{L_f} \neq 0$. Since the coordinate ring of L_f is a domain, $F_f = DG$ is nonzero on L_f, vanishes on both components, and has the asserted degree. It is defined over $\mathbb{F}_q$.

Over an algebraic closure, the ordered root-product map on affine cones is surjective onto $\mathrm{Sym}^m E^\vee$, and the linear catalecticant map is surjective onto the affine cone over L_f. Hence the pullback of the nonzero restriction $F_f|_{L_f}$ is nonzero. Every coefficient of $\lambda_1 \cdots \lambda_m$ is linear in each marker; therefore composition preserves the degree bound separately in every marker. □

Theorem V.4 (Simultaneous-marker escape)**.** Let E be defined over $\mathbb{F}_q$, let $r \geq 6$, let $A \subseteq \mathbb{P}^1(\mathbb{F}_q)$ have size s, and put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus A$ and

$$Q^*_{r,s} = 6(r+s) - 16 + \left\lfloor 2\sqrt{6(r+s) - 18} \right\rfloor .$$

If $q \geq Q^*_{r,s}$, $0 \neq f \in \Gamma^{r-1}E$, and $[f] \notin \mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}$, then W_f contains a split squarefree form of degree $r-2$ supported on S. In characteristic two it is enough that

$$q \geq 6(r+s) - 22 + \left\lfloor 2\sqrt{6(r+s) - 24} \right\rfloor .$$

Proof. Choose a retained point of S as infinity by a projective coordinate change. Then every deleted point lies in the affine marker chart. Lemma V.3 and Lemma V.2 give a split squarefree marker form R, disjoint from A, for which $b = \iota_R f$ lies outside both components of the terminal carrier. The required selector inequality is $q > r+s$, or $q > r+s-2$ in characteristic two; it is weaker than the displayed hypotheses.

Because $b \notin V(D) = \mathcal{P}_5$, Lemma IV.1 makes W_b base-point-free. Proposition IV.3 then places its cubic map in the separable S_3 case. Proposition VII.6 and its branch budget give at least

$$\frac{q + 1 - 2\sqrt{q} - B_p}{6}, \qquad B_p = \begin{cases} 6, & p = 2, \\ 12, & p \neq 2, \end{cases}$$

split squarefree pencil members. A prescribed rational root occurs in at most one member of a base-point-free pencil. Hence a split cubic avoids the $m+s$ roots of R and A whenever

$$q + 1 - 2\sqrt{q} > B_p + 6(m+s).$$

For an integer $\Delta \geq 0$, the least integer bound forced by $q + 1 - 2\sqrt{q} > \Delta$ is

$$\Delta + 2 + \lfloor 2\sqrt{\Delta} \rfloor.$$

Taking $\Delta = 6(r+s) - 18$, or $6(r+s) - 24$ in characteristic two, gives the stated thresholds. Proposition V.1 multiplies the terminal cubic by R, producing the required locator in W_f. □

The proof yields many locators rather than one isolated witness.

Corollary V.5 (Split-witness abundance)**.** Fix r, s. As q tends to infinity through prime powers satisfying the hypotheses of Theorem V.4, every syndrome outside the two carriers has at least

$$\frac{q^{r-4}}{(r-2)!} - O_{r,s}(q^{r-9/2})$$

distinct projective split squarefree members of W_f supported on S, equivalently distinct unordered root sets. This denotes a lower bound, not an asymptotic equality.

Proof. On the affine marker chart, the selector has total degree at most $6m$. After deleting diagonals and forbidden values, the number of good ordered marker tuples is at least

$$(q-s)_m - 6mq^{m-1},$$

where $(q-s)_m = (q-s)(q-s-1)\cdots(q-s-m+1)$, with $6m$ replaced by $4m$ in characteristic two. Each has at least

$$\max\left(0, \left\lceil \frac{q+1-2\sqrt{q}-B_p}{6} \right\rceil - m - s\right)$$

terminal cubics. A degree-$(m+3)$ locator has at most $(m+3)!/6$ ordered marker–cubic decompositions. Division by this multiplicity gives the displayed leading term and error. □

For fixed r, successive specialization of the selector gives a deterministic construction using at most $(r-5)q$ partial-substitution tests and $q+1$ terminal pencil tests, assuming explicit coefficient access. The dense selector has size exponential in r; this is not a uniform polynomial-time decoding theorem.

The next section combines this escape theorem with the explicit carrier and the independent covering-radius inputs to classify the two highest coset-weight shells.

VI. High-weight cosets and code extensions

We now turn the geometric theorem into its coding payoff: the two highest coset-weight shells, equivalently all one-column extensions with Singleton defect at most one, followed by family-wise minimum-support counts. Deleting even one evaluation point changes the answer structurally. The omitted curve points create a weight-r shell, while interiors of split secants incident with the deleted set join tangent and conjugate-secant points in weight $r-1$; the placement of the deletions and the GRS multipliers do not affect the counts.

The internal input is the carrier-and-escape theorem: outside $\mathcal{P}_r \cup \mathcal{M}_{r,p}^{\max}$, there is a degree-$(r-2)$ split squarefree locator supported on S. Promotion to the top shell uses the Seroussi–Roth–Dür radius and extension results, while the full-support rank-two weights come from [5]. What follows is the prescribed-deletion synthesis, its exact counts, and its MDS, NMDS, and enumerator consequences.

A. *Coset weights*

Let the columns of H_S be $w_a\nu_{r-1}(a)$, $a \in S$, with every $w_a \neq 0$. Rescaling an error value absorbs w_a, so

$$d_S(f) = \min\bigl\{|T| : T \subseteq S, \quad f \in \langle \nu_{r-1}(T)\rangle\bigr\}. \tag{11}$$

Thus arbitrary GRS multipliers change neither coset weight nor the projective syndrome geometry. The Hankel criterion identifies a degree-$(r-2)$ split squarefree member of W_f, supported on S, with a representation in (11) of size $r-2$.

Proof of Theorem I.1. The pointed simultaneous-marker theorem gives $d_S(f) \le r-2$ outside $\mathcal{P}_r \cup \mathcal{M}_{r,p}^{\max}$, with the two stated thresholds. If $p > r-1$, no two adjacent entries of Pascal row $r-2$ vanish, so the Lucas carrier is empty.

The rational points of $\mathcal{P}_r$ have four disjoint strata: curve points, interiors of rational split secants, off-curve tangent points, and rational points of conjugate secants. When $s=0$, the Seroussi–Roth–Dür radius gate gives radius $r-1$. Curve points and split-secant interiors have weights one and two, while the full-support rank-two calculation [5, Thms. I.5–I.6] gives weight $r-1$ on every off-curve tangent point and rational point of a conjugate secant. These are therefore exactly the maximum-weight strata.

Suppose $s > 0$. If $a \in A$, the arc property says that $\nu_{r-1}(a)$ is outside the span of every $r-1$ retained columns, while any r retained columns span the syndrome space. Hence $d_S(\nu_{r-1}(a)) = r$. Seroussi and Roth's unique-extension theorem [1, Thm. 1] applies when

$$2 \le r \le |S| - \left\lfloor \frac{q-1}{2} \right\rfloor.$$

Indeed $q \ge Q^*_{r,s}$ implies $q \ge 2(r+s)-3$, and hence $r \le (q+3)/2 - s \le |S| - \lfloor (q-1)/2 \rfloor$. The theorem therefore says that these omitted curve points are all the weight-r directions.

Deleting columns cannot lower the weight of a tangent or conjugate-secant direction. Its full-support weight is $r-1$, and it is not one of the weight-r directions, so its S-weight remains $r-1$. Let f be an interior point of a rational secant with at least one endpoint in A. A representation of f by at most $r-2$ retained columns, combined with the secant representation, would give a nontrivial dependence among at most r distinct normal-rational-curve points. This contradicts the arc property. Hence $d_S(f) \ge r-1$. Since f is not an omitted curve point, the weight-r classification gives $d_S(f) \le r-1$, and therefore equality. Retained curve points have weight one, and interiors of secants whose two endpoints are retained have weight two. The escape theorem deals with every point outside $\mathcal{P}_r$, so the list is exhaustive.

Tangent and conjugate-secant directions contribute $q(q+1)^2/2$. The number of rational secants incident with A is

$$\binom{q+1}{2} - \binom{q+1-s}{2} = \frac{s(2q+1-s)}{2},$$

and each has $q-1$ interior rational points. This gives N_{r-1}. Syndrome vectors parametrize cosets, and every nonzero projective direction has $q-1$ representatives, giving the literal coset counts. □

For the full affine support $A = \{\infty\}$, the theorem reads

$$N_r = 1, \qquad N_{r-1} = \frac{q(q^2 + 4q - 1)}{2}.$$

The unique weight-r direction is classical in this high-rate range; the complete weight-$(r-1)$ shell is the point-deleted conclusion proved here. By the next corollary, the former gives the unique projective MDS-extension direction, up to scaling the appended coordinate, and the latter gives the full NMDS-extension family.

B. MDS and NMDS appended columns

For a nonzero syndrome column f, put

$$\widehat{C}_f = \ker[H_S \mid f].$$

Corollary VI.1 (One-column extensions)**.** For every $f \neq 0$,

$$d(\widehat{C}_f) = d_S(f) + 1. \tag{12}$$

Under the hypotheses of Theorem I.1, with $p > r - 1$, the omitted curve points give exactly the MDS extensions; the tangent, conjugate-secant, and A-incident split-secant points give exactly the NMDS extensions; every other appended column has Singleton defect at least two. For $s = 0$, the MDS class is empty.

Proof. The old columns of H_S have no dependent set of at most r columns. A smallest dependence in $[H_S \mid f]$ either uses only old columns and has size at least $r + 1$, or contains f; after deleting f, the latter is a smallest old-column span of f. This proves (12).

If $d_S(f) = r$, then (12) reaches the Singleton bound $r + 1$, so the extension is MDS. Conversely, an MDS extension forces $d_S(f) = r$.

If $d_S(f) = r - 1$, the primal extension has distance r, one below the Singleton bound. Choose $r - 1$ old columns spanning f. A nonzero functional vanishing on their hyperplane also vanishes on f, and hence gives a dual word of weight at most $|S| + 1 - r$. Its restriction to the old coordinates is a nonzero word in a GRS code of distance $|S| - r + 1$, which is the reverse bound. Thus its weight is exactly $|S| + 1 - r$, and the extension is NMDS. Conversely, an NMDS extension has primal distance r, so (12) forces $d_S(f) = r - 1$. The theorem now gives the stated classes. $\square$

C. Family-aggregate NMDS enumerators

Put $m = r - 1$, $N = |S| = q + 1 - s$, and define

$$\mu_S(f) = \#\{T \subseteq S : |T| = m, \quad f \in \langle \nu_m(T) \rangle\}.$$

For a weight-$(r-1)$ projective syndrome direction $[f]$, every such support gives one minimal dependence, unique up to a nonzero scalar. Thus

$$A_r(\widehat{C}_f) = (q - 1)\mu_S(f). \tag{13}$$

Let Tan, Conj, and Split_A denote the sets of projective tangent, conjugate-secant, and deleted-point-incident split-secant directions in this shell. Sums use one representative of each projective direction. Put $M_{\mathcal{F}} = |\mathcal{F}|$. Explicitly,

$$M_{\mathrm{Tan}} = q(q + 1), \qquad M_{\mathrm{Conj}} = \frac{q(q^2 - 1)}{2}, \qquad M_{\mathrm{Split}_A} = \frac{(q - 1)s(2q + 1 - s)}{2}.$$

Here the last factor $q - 1$ counts interior projective points on each A-incident rational secant, not scalar syndrome representatives.

Theorem VI.2 (Family-aggregate NMDS enumerators)**.** Under the hypotheses of Theorem I.1, with $p > r - 1$,

$$\sum_{f \in \mathrm{Tan}} \mu_S(f) = (q + 1 - m)\binom{N}{m}, \tag{14}$$

$$\sum_{f \in \mathrm{Conj}} \mu_S(f) = \frac{q(q - 1)}{2}\binom{N}{m}, \tag{15}$$

$$\sum_{f \in \mathrm{Split}_A} \mu_S(f) = \left[\binom{s}{2} + s(N - m)\right]\binom{N}{m}. \tag{16}$$

For a family $\mathcal{F}$ above, put $B_{\mathcal{F}} = \sum_{[f] \in \mathcal{F}} \mu_S(f)$, let $L = N + 1$, $K = L - r$, and set $\mathcal{A}_j^{\mathcal{F}} = \sum_{[f] \in \mathcal{F}} A_j(\widehat{C}_f)$. Then

$$\mathcal{A}_0^{\mathcal{F}} = M_{\mathcal{F}}, \qquad \mathcal{A}_j^{\mathcal{F}} = 0 \ (1 \leq j < r), \qquad \mathcal{A}_r^{\mathcal{F}} = (q - 1)B_{\mathcal{F}},$$

and, for $1 \le i \le K$,

$$\mathcal{A}^{\mathcal{F}}_{r+i} = M_{\mathcal{F}} \binom{L}{K-i} \sum_{j=0}^{i-1} (-1)^j \binom{r+i}{j} (q^{i-j} - 1) + (-1)^i \binom{K}{i} (q-1) B_{\mathcal{F}}. \tag{17}$$

Thus the displayed data determine the complete family-aggregate NMDS weight enumerators and, when $M_{\mathcal{F}} > 0$, the family averages. They do not assert that individual extensions in one family have equal weight distributions.

Proof. For $T \in \binom{S}{m}$, let H_T be the hyperplane spanned by its normal- rational-curve points. On the tangent at a, H_T meets the tangent at the curve point if $a \in T$, and at one off-curve point otherwise. Thus each T contributes $q+1-m$ tangent incidences. Every conjugate secant meets H_T in exactly one rational point, giving $q(q-1)/2$ incidences. A secant through two omitted points contributes for every T; a secant through one omitted and one retained point contributes exactly when the retained endpoint is outside T. These give respectively $\binom{s}{2}$ and $s(N-m)$ incidences per T. Summing over all $\binom{N}{m}$ sets T proves the three formulas.

Equation (13) gives $\mathcal{A}^{\mathcal{F}}_r = (q-1)B_{\mathcal{F}}$. The standard NMDS recurrence [12, Thm. 10] expresses every coefficient as

$$A_{r+i}(\widehat{C}_f) = \binom{L}{K-i} \sum_{j=0}^{i-1} (-1)^j \binom{r+i}{j} (q^{i-j} - 1) + (-1)^i \binom{K}{i} A_r(\widehat{C}_f), \qquad 1 \le i \le K.$$

Summing this identity over the $M_{\mathcal{F}}$ directions in a family proves (17) and the final assertion. ☐

VII. Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

a) The criterion in finite-geometric form: At $r = 5$ the witness system $W_f \subset \mathrm{Sym}^3 E^\vee$ is a pencil of binary cubics, that is a line of $\mathrm{PG}(3,q)$ relative to the twisted cubic; and a completely split squarefree cubic is a point of the $\mathrm{PGL}_2(q)$-orbit $\mathcal{O}_3 = \mathrm{PGL}_2(q) \cdot XY(X-Y)$ of size $(q^3-q)/6$, equivalently a plane meeting the twisted cubic in three distinct rational points [6, Prop. 4.1 and Rem. 4.2], [13, Lem. 4.1]. Corollary III.2 therefore reads

$$f \text{ split-free} \iff W_f \cap \mathcal{O}_3 = \varnothing. \tag{18}$$

In this form the redundancy-five question is a point-orbit distribution question for line orbits of $\mathrm{PG}(3,q)$, and as such it has been answered in the finite-geometry literature; the relation is set out in Remark VII.14. What that literature does not supply, and what this section adds, is the passage from a redundancy-five syndrome to its pencil: which lines arise as some W_f, the nonsurjectivity of that image, which *syndromes* are thereby split-free, and how this pencil becomes the terminal engine of the arbitrary-redundancy theorem.

This section therefore has two roles. The arbitrary-redundancy theorem above uses only the exact split-member count and branch budget proved here; the full R5 theorem is a sharper terminal classification placed afterward. Its case map is

$$\begin{aligned} \text{quadratic gcd} &\longrightarrow \mathrm{T}, \mathrm{S}, \\ C_3\text{-cover} &\longrightarrow \mathrm{O}^{\pm},\ \text{nucleus/wild}, \\ S_3\text{-cover} &\longrightarrow \text{small-field sporadics}. \end{aligned}$$

The cyclic and wild families belong to this terminal analysis and are not induction hypotheses for the arbitrary-redundancy result.

Theorem VII.1 (Complete redundancy-five classification)**.** Let $q \ge 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;
(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;
(iii) if $\mathrm{char}\,\mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;

(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;
(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table IV, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\operatorname{char} \mathbb{F}_q = 3$.

Proposition VII.2 (Radius gate)**.** For every prime power $q \geq 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. The Seroussi–Roth extension theorem [1, Thm. 1], applied to the dual five-dimensional GRS code, gives nonextendability when

$$5 \leq \left\lfloor \frac{q}{2} \right\rfloor + 2,$$

apart from its even-field dimension-three exception. The inequality holds for $q \geq 7$, and the exception does not occur. Dür's completeness–covering-radius equivalence [2], as stated by Kaipa [3, Sec. IV and Prop. 4(1)], now gives radius four. Thus Corollary III.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

A. Gcd strata

Proposition VII.3 (Quadratic gcd)**.** If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely, every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma III.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [3, Prop. 1] and [5, Thms. I.4–I.6]. □

B. The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition VII.4 (Cubic incidence and off-diagonal fiber square)**.** The curve X_f is geometrically integral of bidegree $(1,3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2,2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0,e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1,1)$ from the fiber square leaves bidegree $(2,2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. □

Lemma VII.5 (Cyclic stratum)**.** Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0:0:1:0:0)$, whose pencil is $\langle T^3, U^3\rangle$; this fixed point is deep.
(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2\rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2-1)/2$.

Proof. Tame characteristic. Assume that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3\rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. In that case any unramified fiber containing a rational point is its three-point rational deck orbit, and such split fibers exist; without descent no unramified fiber is completely split. Thus the rational-pair pencil is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent, and hence the split-fiber congruence, is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

The fixed wild point. Suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0, 0, 1, 3t, 6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3\rangle$, so it is deep.

The remaining wild orbit. A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2\rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \qquad W_{f_a} = \langle U^3, T^3 + aTU^2\rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. □

The count of split members of the pencil is not merely positive or zero: it is determined exactly by the fiber square, and the following identity is what drives both the threshold and the shape of the exceptions.

Proposition VII.6 (Exact split-witness count)**.** Let $\deg \gcd(W_f) = 0$ and let ϕ_f be separable; the characteristic is arbitrary. Write

- N_f for the number of completely split squarefree members of W_f,
- d_2 for the number of members with a rational double root and a distinct rational simple root,
- d_3 for the number of members that are perfect cubes of a rational linear form.

Then

$$\#Y_f(\mathbb{F}_q) = 6N_f + 3d_2 + d_3. \tag{19}$$

In particular f is split-free if and only if $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$.

Proof. Each $x \in \mathbb{P}^1(\mathbb{F}_q)$ lies on exactly one member g_x of the pencil, because the pencil is base-point-free; write $g_x = \ell_x h_x$ with ℓ_x the linear form at x. The fiber of Y_f over x under the first projection is the set of rational roots of h_x, so $\#Y_f(\mathbb{F}_q) = \sum_x \#\{\text{rational roots of } h_x\}$, and it suffices to total that sum member by member. A member with three distinct rational roots is met at each of them, and there h_x has the two remaining roots, contributing $3 \cdot 2 = 6$. A member with a rational double root r and distinct rational simple root s is met at both: over r the residual is $\ell_r \ell_s$ with two distinct rational roots, over s it is ℓ_r^2 with one, contributing $2 + 1 = 3$. A perfect cube is met only at its root, where the residual is a square, contributing 1. A member with one rational root and an irreducible quadratic cofactor contributes 0 there, and a member with no rational root is met at no rational x. Summing gives (19). □

Lemma VII.7 (Branch budget)**.** Under the hypotheses of Proposition VII.6,

$$d_2 + 2d_3 \le 4,$$

so a split-free f has $\#Y_f(\mathbb{F}_q) \le 12$, with equality only when $d_2 = 4$ and $d_3 = 0$.

Proof. Riemann–Hurwitz gives different degree four for the separable degree-three map ϕ_f. A member with a double root has one ramification point of index two; it spends different degree one when that ramification is tame and at least two when it is wild, which happens exactly in characteristic two. We avoid the term *simple ramification*, which is usually taken to include the tame exponent. A member that is a perfect cube is totally ramified; it spends two when tame and at least three when wild, which happens exactly in characteristic three. Distinct members are distinct branch values, so $d_2 + 2d_3 \leq 4$ in every characteristic, and characteristic two gives the sharper $d_2 + d_3 \leq 2$. Substituting into $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$ and maximizing over the integer points of that region gives 12, attained only at $(d_2, d_3) = (4, 0)$. □

Corollary VII.8 (Split members of an S_3 pencil)**.** If ϕ_f has geometric monodromy S_3, then

$$\frac{q + 1 - 2\sqrt{q} - B}{6} \;\leq\; N_f \;\leq\; \frac{q + 1 + 2\sqrt{q}}{6},$$

where $B = 12$, and $B = 6$ when $\operatorname{char}\mathbb{F}_q = 2$.

Proof. Proposition VII.6 gives $6N_f = \#Y_f(\mathbb{F}_q) - 3d_2 - d_3$, and Lemma VII.7 bounds $3d_2 + d_3$ above by 12 in general and by 6 in characteristic two, where $d_2 + d_3 \leq 2$; it is bounded below by 0 in every characteristic. The S_3 hypothesis makes Y_f geometrically integral of arithmetic genus one, so Aubry–Perret confines $\#Y_f(\mathbb{F}_q)$ to $q + 1 \pm 2\sqrt{q}$ [14, p. 468]. □

Corollary VII.9 (Binary S_3 shallowness above $q = 8$)**.** If $\operatorname{char}\mathbb{F}_q = 2$ and ϕ_f has geometric monodromy S_3, then f is shallow for every $q \geq 16$.

Proof. A split-free f has $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3 \leq 6$ by Lemma VII.7, while Aubry–Perret gives $\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$. For $q = 16$ the latter is 9, and it increases with q, so no binary field with $q \geq 16$ admits a split-free S_3 pencil. □

The identity class has S_3 splitting density $1/6$, so $(q + 1)/6$ is the expected count. A completely split fiber is one whose Frobenius class is the identity, and that class has density $1/|S_3| = 1/6$ on the unramified locus, by the factorization-type correspondence of [15, Thm. 3.6] together with the Chebotarev count and density of [15, Thm. 3.7 and Cor. 3.8], whose geometric Galois group is S_3 under the present hypothesis; so $(q + 1)/6$ is a Chebotarev main term rather than a count of fibers. Corollary VII.8 makes it quantitative: split members occur with that expected density up to a Weil-scale error and a bounded ramification correction. Split-freeness is the extreme small-field case in which the lower bound has turned nonpositive, which is why the exceptional fields are small ones. Lemma VII.10 records only the sign of this statement. We claim the density only for the completely split class; the other factorization types are not counted here.

Lemma VII.10 (S_3 stratum)**.** If ϕ_f has geometric monodromy S_3 and $q \geq 20$, then the pencil contains a completely split squarefree cubic. For odd q the same conclusion holds for every $q \geq 20$ without the monodromy hypothesis whenever Y_f is geometrically integral.

Proof. By Proposition VII.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. The Aubry–Perret singular-curve bound, in the form

$$\bigl|\#C(\mathbb{F}_q) - (q + 1)\bigr| \leq 2\pi_C\sqrt{q}$$

for a reduced geometrically integral projective curve of arithmetic genus π_C, gives

$$\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$$

[14, p. 468]. Suppose f were split-free. Then $\#Y_f(\mathbb{F}_q) \leq 12$ by Lemma VII.7, so $q + 1 - 2\sqrt{q} \leq 12$, that is $(\sqrt{q} - 1)^2 \leq 12$ and $q \leq (1 + 2\sqrt{3})^2 < 20$. □

Corollary VII.11 (Forced fiber-square invariants)**.** Let q be odd with $17 \leq q \leq 19$ and let f be split-free with Y_f geometrically integral. Then

$$\#Y_f(\mathbb{F}_q) = 12, \qquad d_2 = 4, \qquad d_3 = 0.$$

Proof. Lemma VII.7 restricts $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$ to the values $0, 1, 2, 3, 4, 6, 7, 9, 12$. Aubry–Perret forces $\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$, which exceeds 9 for $q \geq 17$; the only admissible value left is 12, and by Lemma VII.7 that value is attained only at $(d_2, d_3) = (4, 0)$. □

Proposition VII.12 (Exhaustion and finite bridge)**.** The preceding cases exhaust every redundancy-five pencil for $q \geq 20$, and in characteristic two for $q \geq 16$ by Corollary VII.9. The finite residue is therefore

$$q \in \{7, 8, 9, 11, 13, 17, 19\},$$

which Certificate R5 closes. The certificate additionally covers $q = 16$, where it finds no sporadic orbit; that field is no longer logically certificate-dependent, and the computation is retained as an independent regression check. Sporadic S_3-stratum orbits occur exactly at $q = 7, 8, 9, 11, 13, 17, 19$.

Proof. If H_f has rank one, then f is a normal-rational-curve point and $W_f = L\,\mathrm{Sym}^2 E^\vee$, which contains a split squarefree cubic. Suppose therefore that H_f has rank two. By Lemma IV.1, a nontrivial gcd has degree two, and Proposition VII.3 settles that case. Thus the remaining pencil has trivial gcd, and its cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas VII.5 and VII.10. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. ☐

The fiber square is also the genus-one curve of the incidence literature.

Proposition VII.13 (The fiber square is the incidence curve)**.** Keep the hypotheses of Proposition VII.6, assume $\mathrm{char}\,\mathbb{F}_q \neq 2$. For each $x \in \mathbb{P}^1$, let g_x be the unique pencil member vanishing at x, write $g_x = \ell_x h_x$, and put $D_f(x) = \mathrm{disc}(h_x)/4$. The restriction is needed here and not in Proposition VII.6: in characteristic two the quotient by four is undefined, and a separable quadratic cover is Artin–Schreier rather than the Kummer cover displayed below. The first projection realizes Y_f as the double cover

$$w^2 = D_f(x)$$

of $\mathbb{P}^1$, branched where D_f vanishes. If moreover W_f misses the twisted cubic and $\mathrm{char}\,\mathbb{F}_q \neq 2, 3$, then D_f is the discriminant quartic that Kaipa and Pradhan attach to the line, so the normalization of Y_f is their elliptic curve E_{W_f} and

$$\#Y_f(\mathbb{F}_q) = \#E_{W_f}(\mathbb{F}_q).$$

Proof. The fiber of Y_f over x under the first projection is the root set of the residual quadratic h_x, and its two roots are interchanged by the choice of a square root of $\mathrm{disc}(h_x)$; so the projection is the stated double cover, and it is branched exactly where that discriminant vanishes. Kaipa and Pradhan form the discriminant of the same residual quadratic and define E_L as the nonsingular model of $w^2 = D_L$ [13, §4–§5], taking as we do one quarter of the discriminant of the same residual quadratic. They therefore differ by the square of the nonzero rational function $\lambda \in \mathbb{F}_q(x)^\times$ that relates the two normalizations of the residual quadratic,

$$D_f = \lambda^2 D_L \quad \text{in } \mathbb{F}_q(x)^\times,$$

with equality when the representative of g_x is normalized as they normalize it. Equality of square classes in the function field, rather than pointwise at rational x, is what rules out a quadratic twist and leaves the cover unchanged. For the point counts, note that on that stratum D_f has nonzero discriminant, so the double cover displayed above is already smooth and is its own normalization, and Y_f is smooth there as well; the two smooth projective models are then isomorphic and their counts agree. Away from that stratum the identification is birational and the counts may differ at a rational singularity, which is why the exact count of Proposition VII.6 is stated for $\#Y_f(\mathbb{F}_q)$ rather than for the count on a normalization. ☐

Proposition VII.6 is a counting identity valid in every characteristic, and the quantitative splitting law of Corollary VII.8 follows from it directly. In the tame range it is equivalent to an incidence count in the twisted-cubic line literature. Precisely: when W_f misses the twisted cubic, so that $d_3 = 0$, and $\mathrm{char}\,\mathbb{F}_q \neq 2, 3$, the quantity d_2 is the invariant η_L attached to the binary quartic of the line and $\#Y_f(\mathbb{F}_q)$ is the point count of the associated elliptic curve, so that (19) reads

$$N_f = \frac{\#E_L(\mathbb{F}_q) - 3\eta_L}{6}.$$

This is the incidence count of [13]. We use the form derived in their proof, $|L \cap \mathcal{O}_3| = (\nu_L - \eta_L)/3$ and $\nu_L = (\#E_L(\mathbb{F}_q) - \eta_L)/2$ [13, Prop. 4.5(3) and Thm. 5.1]; the denominator printed in their Theorem 1.3(3) differs from this proof consequence. The denominator 6 also makes the five incidence counts sum to $q + 1$.

Proposition VII.6 itself drops the restrictions on characteristic, includes lines meeting the curve through the d_3 term, and uses neither invariant theory nor a discriminant. Only the elliptic-model comparison remains restricted to characteristic different from two and three. The supplement exhaustively checks (19) and Lemma VII.7 over every point of $\mathrm{PG}(4, q)$ for all prime powers $4 \leq q \leq 32$.

Remark VII.14 (Relation to the twisted-cubic line literature)*.* Two conventions for "the line of $\mathrm{PG}(3, q)$ attached to a pencil of binary cubics" are in use, and they are exchanged by a polarity, so the class names below must be read in the right one. Günay and Lavrauw, and Blokhuis, Pellikaan, and Szőnyi, send a cubic to a *plane*: the twisted cubic is a curve of points, a cubic is the plane meeting it in that cubic's divisor, and a pencil of cubics becomes a pencil of planes, whose axis is the line [16, §2], [6, Prop. 6.5]. Kaipa and Pradhan send a cubic to a *point*, so that the twisted cubic consists of the perfect cubes and the pencil *is* the line [13, §1]. The bridge is the symplectic polarity σ of $\mathrm{PG}(3, q)$, which carries each point of the twisted cubic to its osculating plane and carries the chords of the twisted cubic to the axes of the osculating developable [16, §2].

For us the translation is concrete. The osculating plane at the point λ^3 of the twisted cubic is $\lambda\,\mathrm{Sym}^2 E^\vee$, so

$$\mathrm{Osc}(\lambda_1) \cap \mathrm{Osc}(\lambda_2) = \lambda_1\lambda_2 \langle T, U \rangle$$

is exactly a quadratic-gcd pencil, which is therefore an axis in the point convention; by [16, Lem. 2] its polar is the chord joining λ_1^3 and λ_2^3, which is what the plane convention calls it. Symmetrically $\langle T^3, U^3\rangle$ joins two points of the twisted cubic and so is a chord in the point convention and an axis in the plane convention. The class names in this remark follow the plane convention of the source being cited; the split members of a pencil are of course the same either way.

Under (18) every stratum above is a class of lines of $\mathrm{PG}(3,q)$ in the sense of Blokhuis, Pellikaan, and Szőnyi, who partition the lines into classes $O_1,\dots,O_8$ and determine for $q \geq 23$, in every characteristic, which classes lie in a plane meeting the twisted cubic in three rational points [6, Thm. 8.1 and Prop. 8.4]. This ambient line-class correspondence is exact. Our quadratic-gcd trichotomy of Proposition VII.3 is their degree-one case: real chords O_1 are shallow, tangents O_2 give the tangent family, imaginary chords O_3 give the conjugate-secant family. Their unisecant classes O_4 and O_5, whose pencils have a common factor of degree exactly one, are not realized by a rank-two Hankel syndrome, by Lemma IV.1. The two cyclic cases of Lemma VII.5(i) are their real axes O_1' and imaginary axes O_3', with the same congruences mod three; the characteristic-three nucleus of Lemma VII.5(ii) is their axis $O_7(3)$; the wild pencils of Lemma VII.5(iii) are their $O_{8.1}(3)^{\pm}$ and $O_{8.2}(3)$. Lemma VII.10 is their O_6, and their Remark 7.12 reaches the threshold $q \geq 23$ by the same route: the double point scheme of the degree-three map is a genus-one curve, at most twelve points must be excluded, and the Hasse–Weil bound then produces a fiber with three distinct rational points. Two differences remain. Aubry–Perret in place of Hasse–Weil drops their simplicity hypothesis, which is what lets one estimate carry the singular and wild cases uniformly; and the exact count of Proposition VII.6 replaces the deletion bookkeeping by an identity, which lowers the threshold to $q \geq 20$ and, through Corollary VII.11, pins the fiber square of every remaining exception at $q = 17, 19$. Independent confirmation of the same table comes from the classification of degree-three permutation rational functions [17], as recorded in [6, Rem. 8.6].

Exact counts, rather than the vanishing test, are available too. For the ten non-generic line classes Davydov, Marcugini, and Pambianco give the exact number of planes of each orbit through each line for $q \geq 5$ [18, Thm. 3.3], building on their line-orbit classification [19], [20]; Günay and Lavrauw give the point-orbit and plane-orbit distributions of the same classes for odd q prime to three [16]. For the generic class in characteristic different from two and three, Kaipa and Pradhan's elliptic formula is exactly the specialization of Proposition VII.6 established above, not an independent route to Lemma VII.10. The generic class in characteristics three and two is treated respectively in [21] and [22].

What this section contributes is therefore not the pencil-level classification but its redundancy-five syndrome-level source and consequence. Blokhuis–Pellikaan–Szőnyi formulate the codimension-four generalized Reed–Solomon code in syndrome/coset language and determine its extended coset-leader weight enumerator; their line criterion does not supply the map from a redundancy-five syndrome to the pencil classified here. The passage $f \mapsto W_f$ of Lemma III.1 is not present in that literature, and it is not surjective onto the lines: in particular, no unisecant with gcd of degree one is realized by a rank-two syndrome. Neither do incidence distributions produce syndromes; Table IV records canonical representatives in divided-power syndrome coordinates with their orbit sizes, stabilizers, and Frobenius fusion, and extracting such a list from [13, Prop. 4.5 and Thm. 5.1] would still require evaluating $\#E_L$ and η_L over all $2q-3+\mu$ generic line orbits. The promotion of this redundancy-five syndrome classification from split-free to deep, through Proposition VII.2, is also not supplied by the cited line-incidence classifications.

Redundancy five is the terminal model for the transverse branch of the recursion. Its proof separates those later tasks cleanly: identify the lower splitting curve, remove its singular, branch, diagonal, and marker incidences, apply a rational-point bound, and read a surviving point as a split witness. Higher redundancy changes how one reaches that curve and which polar lines remain trapped. The general contained branch is classified by Theorem IV.7; Appendix B resolves the first two fixed levels separately.

TABLE IV: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \ldots, a_4]$ is in the divided-power syndrome coordinates of Section III. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	`[0,1,0,0,3]`	28	12	–
7	`[0,1,0,0,2]`	112	3	–
7	`[0,0,1,0,3]`	168	2	–
7	`[0,1,0,3,2]`	168	2	–
7	`[1,0,0,2,1]`	168	2	–
8	`[0,1,1,0,2]`	252	2	$\to [0,1,1,0,4]$
8	`[0,1,1,0,4]`	252	2	$\to [0,1,1,0,6]$
8	`[0,1,1,0,6]`	252	2	$\to [0,1,1,0,2]$
9	`[1,0,0,1,2]`	180	4	–
9	`[0,1,0,4,1]`	360	2	$\leftrightarrow [0,1,0,4,4]$
9	`[0,1,0,4,4]`	360	2	$\leftrightarrow [0,1,0,4,1]$
11	`[1,0,0,1,10]`	330	4	–
11	`[1,0,0,1,1]`	660	2	–
13	`[0,1,0,0,4]`	182	12	–
13	`[1,0,0,4,7]`	546	4	–
17	`[1,0,0,1,5]`	1224	4	–
19	`[0,1,0,0,4]`	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table IV are therefore visibly distinguished without requiring a working-directory file.

TABLE V

INDEPENDENT R5 COMPLETENESS TOTALS. THE FIRST EQUALITY IS THE DIRECT SYNDROME ENUMERATION VERSUS THE ORBIT–STABILIZER SUM; THE SECOND DECOMPOSES THE SAME TOTAL INTO THE FORMULA IN THEOREM VII.1 AND THE SPORADIC ROWS ABOVE.

q	direct $=$ orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

Remark VII.15 (No one-column MDS extension at this radius). At $q = 8$ the split-free count of Table V is 1116, splitting as 360 nonsporadic and 756 sporadic directions. These are deep holes of $\mathrm{PRS}_{\mathbb{F}_8}(4)$ by Proposition VII.2 and Corollary III.2; they are not one-column MDS extensions, and no such extension exists.

The two notions differ by one column. A syndrome direction is a deep hole when it lies in the span of no $r-2$ parity-check columns and the radius is $r-1$; adjoining a point to the $(q+1)$-arc of the normal rational curve keeps it an arc, equivalently extends the dual code by one column, only when that point lies in the span of no $r-1$ columns. At redundancy five the first condition gives distance at least four and Proposition VII.2 gives radius exactly four, so no direction achieves the distance five that an extension requires. Dür's equivalence states this as a biconditional: the covering radius is $r-1$ precisely when the arc of every normal rational curve in $\mathrm{PG}(r-1, q)$ is complete, so there is no MDS extension of the dual by one digit [2], as stated by Kaipa [3, Sec. IV]. The same reading applies at every redundancy classified here, since each radius gate proves $\rho = r - 1$.

VIII. SHARP FIXED-LEVEL AND MODULAR REFINEMENTS

Proposition IV.7 identifies every component on which contraction can remain trapped, and Theorem V.4 handles its complement uniformly. The retained fixed levels give exact classifications below the uniform field threshold and resolve the first modular carriers. Their proofs use stagewise polar packages, marker and ramification divisors, and explicit adjacent-zero Hankel kernels. These calculations refine the general theorem; they are not its induction steps.

The exceptional-orbit and split-free classifications at redundancies six and seven are the fixed-level geometric contribution, stated in Theorems B.9 and B.14. Their proofs separately supply the covering-radius promotion; Proposition IV.10 concerns the higher fixed levels and the general large-characteristic range.

Theorem VIII.1 (Redundancy-six and redundancy-seven deep holes)**.** The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the five finite-field exception rows of Theorem B.9, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.

(b) For every prime power $q \geq 7$, Theorem B.14 gives the complete split-free redundancy-seven classification. For $q \geq 13$ only the persistent families and, when $q = 2^m$ with odd m, one central nucleus point remain. The result is a deep-hole classification for every $q \geq 7$. The radius is six except at $q = 8$, where it is seven; exact extraction leaves the nine-direction diagonal tangent orbit and the fixed central nucleus as the ten deep directions.

At redundancies six and seven, the first two pointed packages expose the stagewise mechanism in its smallest dimensions. The persistent rank-two carrier, its rank-one boundary, the modular nucleus, and the self-collision divisor account for every polar line that can fail to escape. The remaining line meets a proper bad scheme of bounded degree, and the terminal cubic cover supplies a split squarefree witness. Appendix B proves the exhaustive contained-component classification and closes the bounded fields by finite certificates. Higher fixed-level calculations are not needed for the arbitrary-redundancy theorem and are left to the companion record.

IX. Scope and open problems

The recursive carrier theorem closes the reduced contained-component problem at arbitrary redundancy. The remaining arbitrary-level obstruction is the split-squarefree arithmetic of later maximal Lucas carriers in small characteristic.

1) Redundancy five is complete for every $q \geq 7$.
2) Redundancy six is a complete deep-hole classification over every prime power $q \geq 7$. Redundancy seven has a complete deep-hole classification over the same range; its radius is six except at $q = 8$, where its radius is seven and exact extraction gives ten deep directions in the diagonal tangent and central-nucleus orbits.
3) The exact R8–R10 calculations belong to the companion record and are not claims of this submission.
4) At arbitrary redundancy, the reduced carrier and its degree-one modular union are exact. Split-squarefree arithmetic on later Lucas carriers remains open outside the evaluated levels.

In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture. It proves the reduced carrier and split-free containment under the displayed field-size bound. When the characteristic exceeds $r-1$, the Lucas carrier is empty and the separate coding-theoretic radius gate promotes that containment to the exact high-weight-coset classification.

The last sporadic fields at redundancies five, six, and seven are $19, 13, 11$, while the corresponding first prime-power field-order gates are $23, 29, 37$. These opposite trends ask whether sufficiently high redundancy has any sporadic fields outside the persistent and modular loci.

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Appendix A
Stagewise interfaces for the fixed-level refinements

The arbitrary-redundancy theorem uses simultaneous contraction, not the stagewise statements below. The retained R6–R7 appendix uses them to obtain sharper small-field bounds and isolate the first exceptional modular components; the companion R8–R10 calculations are not claims of this submission. This section records the interfaces needed here, with proofs, so that the submission is self-contained.

For $\lambda \in E^\vee$, divided-power contraction satisfies

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{20}$$

The lifted form is squarefree precisely when g is squarefree and avoids the marked root λ.

Theorem A.1 (Polar-flag construction)**.** Iterated contraction commutes with field extension and $\mathrm{GL}(E)$. If $g \in W_{\iota_{\lambda_j} \cdots \iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$. If the projective markers $\lambda_1, \ldots, \lambda_j$ are pairwise distinct, the lift is squarefree exactly when g is squarefree and avoids every marker.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (20) gives membership. The product of the distinct markers is squarefree, and its product with g is squarefree exactly when g is squarefree and shares no marker factor. $\square$

Definition A.2 (One-step lower package)**.** A one-step lower package consists of a closed lower bad locus, finitely many geometric strata on its complement, a proper geometrically integral splitting curve on each stratum with stated genus and rational-point bound, a zero-dimensional deletion scheme containing branch, diagonal, collision, and marker incidences, and a finite parameter scheme containing every unavailable marker. Every rational point outside the deletion scheme supplies a split squarefree lower witness avoiding all retained markers.

Definition A.3 (Contained-component assertion)**.** For a lower bad scheme B and a closed upper list $\mathcal{C}$, the contained-component assertion says that every syndrome whose contraction is noninjective, whose entire polar line lies in B, or whose collision divisor is the whole marker line belongs to $\mathcal{C}$.

Lemma A.4 (Collision criterion)**.** If W_f has trivial gcd, then a marker λ is a repeated factor in every lift through λ if and only if λ is a fixed factor of $W_{\iota_\lambda f}$; equivalently, it lies on the collision divisor of the linear series W_f.

Proof. Equation (20) gives

$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee.$$

After division by λ, repeated lifting is therefore equivalent to a fixed lower factor. Since W_f is base-point-free, the same condition says that the morphism defined by W_f has zero differential at λ, which is the collision condition. $\square$

Lemma A.5 (Uniform collision bound)**.** For $j \geq 5$, the collision divisor of a base-point-free g^{j-4}_{j-2} on $\mathbb{P}^1$ has degree at most $2j-6$ unless the induced morphism is inseparable. The only inseparable numerical cases are $(j,p) = (5,3)$ and $(6,2)$.

Proof. An inseparable morphism factors through absolute Frobenius. Hence $p \mid (j-2)$, while its $j-3$ sections lie in the $(j-2)/p+1$-dimensional space of p-th powers; thus $p(j-4) \leq j-2$. This leaves only $(j,p) = (5,3),(6,2)$. In the separable case a nonzero value–derivative minor is a binary Wronskian of degree $2(j-2)-2 = 2j-6$ containing the collision scheme. $\square$

Theorem A.6 (Finite-depth polar escape)**.** Suppose s coherent contraction stages have unavailable-marker schemes of degrees $d_i < q+1$, each containing every previously chosen marker. Suppose each surviving terminal stratum has a proper geometrically integral curve with

$$\#Y(\mathbb{F}_q) \geq q + \kappa - 2g\sqrt{q}$$

and at most δ rational points that fail to supply a split squarefree terminal witness avoiding all markers. If

$$q + \kappa - 2g\sqrt{q} > \delta,$$

then W_f contains a split squarefree member.

Proof. Choose the markers successively. A zero-dimensional scheme of degree below $q+1$ cannot contain every rational point of $\mathbb{P}^1$, so each stage has a surviving choice. On the terminal stratum the displayed inequality leaves a rational point outside the bad set. It gives a split squarefree terminal member avoiding every marker. Repeated use of (20) multiplies by the pairwise distinct marker factors, and avoidance preserves squarefreeness. $\square$

Proposition A.7 (Contained rank-two polar lines)**.** Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and assume its contraction map is injective. Then the whole first-polar line lies in the lower rank-at-most-two catalecticant locus if and only if f lies in the upper rank-at-most-two locus. Otherwise the rank-drop parameters form a scheme of degree at most three on the marker line.

Proof. Let $A = C_f^{(2)}$ and $K = \ker A$. Multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with $H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee$, and functoriality gives $C^{(2)}_{\iota_\lambda f} = A|_{H_\lambda}$. The forward implication is immediate. Conversely, if $\operatorname{rank} A = 3$, then $\dim K = n-4$; lower rank at most two forces $K \subset H_\lambda$. Containment for every geometric λ would put K in the intersection of all H_λ, which is zero, a contradiction. Finally, the maximal minors of $A|_{H_\lambda}$ are cubics in λ, so in the noncontained case their common zero scheme has degree at most three. $\square$

APPENDIX B
REDUNDANCIES SIX AND SEVEN: COMPLETE AND GATED RESULTS

This appendix takes the marked polar-escape theorem as input and discharges its first two pointed lower packages. The bottleneck is exhaustiveness: the persistent rank-two carrier, modular nucleus, self-collision, and transverse cases must account for every polar line. The resulting structural arguments, together with the bounded certificates, prove Theorems B.9 and B.14.

The numerical mechanism is compact:

level	stages	marker-scheme degrees	terminal bad points	first field gate
R6	1	13	18	29
R7	2	$12, 14$	24	37.

The corresponding contained residues are the persistent carrier together with the binary nucleus line at R6 and its central-point lift at R7.

A. Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition B.1 (Persistent net and orbit law)**.** If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The split-free persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. $\square$

Proposition B.2 (First-polar line and secant degree)**.** For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012} r^3 + \Delta_{013} r^2 - \Delta_{023} r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \le 2$. On the rank-two open this is equivalent to W_f having quadratic gcd; rank one is the normal-rational-curve boundary. In the trivial-gcd case it therefore has at most three zeros. $\square$

Proposition B.3 (Complete lower trivial-gcd carrier)**.** Let $c \in \mathbb{P}(\Gamma^4 E)$ and suppose that the cubic pencil W_c has trivial gcd. One of the following applies over an algebraic closure, and the last three alternatives exhaust the cases in which the S_3 package in (i) is unavailable:

(i) the cubic map is separable with geometric monodromy S_3, and its off-diagonal fiber square is geometrically integral;

(ii) the characteristic is neither two nor three and the pencil is cyclic, in which case c lies on the closure of

$$\left\{\mathrm{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\mathrm{Sym}^2 E^\vee)\right\};$$

(iii) the characteristic is two and c lies on $\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle$;

(iv) the characteristic is three and c lies on $\mathrm{Join}(e_2, C_4)$, the closure of the wild cyclic pencils together with the inseparable nucleus point.

Thus these displayed tame, binary, and ternary loci exhaust the trivial-gcd lower pencils for which the S_3 package is unavailable.

Proof. Proposition VII.4 associates to W_c a degree-three map. If it is separable, its transitive geometric monodromy is C_3 or S_3. In the S_3 case two-transitivity makes the off-diagonal fiber square geometrically integral, giving (i). In the cyclic case, Lemma VII.5 identifies the two ramification points with a binary quadratic Q. Substitution in the two Hankel equations gives the divided-power syndrome $\mathrm{diag}(1,4,6,4,1)^{-1}[Q^2]$ whenever the displayed diagonal is invertible. This proves (ii) in characteristics other than two and three.

The same coefficient calculation can be made without division. In characteristic two it is

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0], \qquad g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

Its image is dense in the plane $\mathbb{P}\langle e_1, e_2, e_3\rangle$, so its closure is exactly the binary locus in (iii). In characteristic three every separable cyclic pencil is wild. Lemma VII.5 puts it in the PGL_2-orbit of $e_2 - ae_4$. Here e_2 is fixed and the orbit closure of e_4 is the quartic normal rational curve C_4; hence the complete wild closure is $\mathrm{Join}(e_2, C_4)$. The only inseparable trivial-gcd pencil is the all-cube nucleus pencil, namely its vertex e_2, by Proposition VII.12. This proves (iv) and exhausts the inseparable case. The three alternatives are therefore exhaustive. □

Proposition B.4 (Cyclic, collision, and terminal bounds)**.** For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled projected quadratic Veronese surface of degree four and contains no polar line;

(ii) the pointed ramification condition has degree at most six and does not vanish identically;

(iii) on an S_3 lower pencil, at most eighteen rational points of the off-diagonal fiber square fail to encode a completely split member avoiding a specified marked root.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{\mathrm{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\mathrm{Sym}^2 E^\vee)\right\}.$$

When 2 and 3 are invertible this is the projected quadratic Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition B.5. For (ii), Lemma A.5 with $j = 6$ gives the degree-six bound except possibly in characteristic two. In that characteristic, inseparability would identify W_f, after a change of coordinates, with the full pure-square space $\langle T^4, T^2U^2, U^4\rangle$. Its annihilator is $\langle e_1^\vee, e_3^\vee\rangle$. For the two consecutive Hankel rows this would require simultaneously

$$a_0 = a_2 = a_4 = 0, \qquad a_1 = a_3 = a_5 = 0,$$

contrary to $f \neq 0$. Thus the bound holds in every characteristic.

Finally, Proposition VII.6 and Lemma VII.7 show that at most twelve rational points encode nonsplit members. Base-point-freeness gives a unique pencil member through the specified marker, and it contributes at most six further points. Thus the terminal bad set has size at most 18, and a permitted split member exists as soon as

$$q + 1 - 2\sqrt{q} > 18.$$

The first prime-power solution is 29. □

Proposition B.5 (Modular contained components)**.** In characteristic three the lower wild carrier is the degree-four cone

$$\mathrm{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. ☐

Proposition B.6 (Binary nucleus arithmetic)**.** The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q+1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

If $C = 0$, the member $A + Bt + Ct^4$ has degree at most one and cannot have four distinct roots. Assume $C \neq 0$. The root differences of $A + Bt + Ct^4$ then lie in the kernel of $u \mapsto u^4 + (B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q - 1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. ☐

Proposition B.7 (Exhaustive R6 contained-component classification)**.** For a redundancy-six syndrome, the alternatives in Definition A.3 are exhausted as follows:

(i) a noninjective contraction map is the rank-one normal-rational-curve boundary and is shallow;
(ii) a polar line contained in the lower secant cubic comes from the upper rank-two catalecticant locus; after its shallow rational secants are removed, this is the persistent tangent and conjugate-secant locus;
(iii) no trivial-gcd polar line is contained in the tame cyclic surface or the characteristic-three wild cone;
(iv) in characteristic two, containment in the cyclic plane occurs exactly for $f \in \mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle$;
(v) on the remaining trivial-gcd locus the collision divisor is not identically zero.

Consequently the complete nonshallow contained list is the persistent locus together with $\mathcal{N}_3$ in characteristic two. There is no unlisted contained component.

Proof. The noninjective and secant-cubic alternatives are Proposition A.7 and Proposition B.2. The tame cyclic and collision assertions are Proposition B.4; the wild and binary specializations are Proposition B.5. Proposition B.3 proves that these are the complete trivial-gcd lower failure loci, rather than a selected list of examples. Together they are all the clauses of the terminal carrier in Definition A.3: contraction indeterminacy, containment in each component of the lower bad scheme, and identical collision. Hence the list is exhaustive. ☐

Proposition B.8 (R6 one-step exhaustion)**.** Assume $q \geq 29$. Then every split-free redundancy-six syndrome lies in the persistent quadratic-gcd locus or, when $q = 2^m$ with odd m, in $\mathcal{N}_3(\mathbb{F}_q)$.

Proof. Contained nets. The complete contained branch is Proposition B.7. Lemma IV.1 puts every common-factor system in the persistent rank-two locus or on its shallow rank-one boundary. Among the rank-two points, removing the shallow rational secants leaves exactly the tangent and conjugate-secant families. It remains to treat a trivial-gcd net.

Transverse exclusions. Its first-polar line is not contained in the lower secant cubic, by Proposition B.2. The pullback of that cubic has degree at most three. The tame cyclic surface has degree four and contains no polar line. In characteristic three its wild replacement has the same degree and no contained trivial-gcd polar line; in characteristic two the replacement is a plane, whose only contained component is $\mathcal{N}_3$, by Proposition B.5. Finally, Proposition B.4 gives a nonzero collision divisor of degree at most six. Outside the contained binary component, the excluded parameter scheme therefore has degree

$$d \leq 3 + 4 + 6 = 13.$$

Bottom pencils. For every remaining marker, the lower syndrome lies outside $\mathcal{P}_5$ and the cyclic locus. Lemma IV.1 makes its cubic pencil base-point-free, and Lemma VII.10 supplies the geometrically integral off-diagonal curve. If no split member avoids the marker, Proposition B.4(iii) bounds its rational points by eighteen. Aubry–Perret therefore supplies an allowed member when

$$q + 1 - 2\sqrt{q} > 18,$$

whose first prime-power solution is 29. Together with the one-step parameter package above, of degree $d = 13$, this terminal bad-point bound meets Theorem A.6, which produces a split squarefree quartic.

The modular residue. This is one field-order threshold, not three characteristic-dependent estimates: the first characteristic-three and characteristic-two prime powers above it happen to be 81 and 32, respectively. On $\mathcal{N}_3$, Proposition B.6 gives the asserted odd-degree split-free orbit and proves shallowness in even extension degree. $\square$

The covering-radius gate is complete. Its exact high-rate endpoint is $q = 8$, since $6 = \lfloor 8/2 \rfloor + 2$. Thus Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 8$, while Certificate R6 checks it directly at $q = 7$ and independently confirms $q = 8$.

Theorem B.9 (Redundancy six)**.** For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent locus of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;
(ii) one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \geq 5$.

The lower bound $m \geq 5$ avoids double counting: the remaining odd case $m = 3$, namely $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table. There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five. At $q = 7, 8, 9, 11, 13$, the exceptional direction totals are

$$5152, \quad 4713, \quad 1800, \quad 792, \quad 546,$$

and the corresponding total numbers of deep syndrome directions are

$$5376, \quad 5037, \quad 2250, \quad 1584, \quad 1820.$$

Proof. Propositions B.7 and B.8 give the complete conceptual exhaustion for every prime power $q \geq 29$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. $\square$

The listed sources are disjoint. The persistent locus has quadratic gcd, every finite exceptional row has trivial gcd, and the separate nucleus term occurs only for binary fields beyond the finite table.

The five small tables are exhaustive computations. For an exceptional net W_f, let $I_f(r)$ be the number of irreducible cubic members of the quotient pencil $W_{b(r)}$, including the infinity chart, and define its shared-root pair count by

$$E_f = \sum_{r \in \mathbb{P}^1(\mathbb{F}_q)} \binom{I_f(r)}{2}.$$

In odd characteristic let

$$F_f(T, U) = \sum_{i=0}^{5} a_i T^{5-i} U^i$$

be the binary quintic attached to $f = (a_0 : \cdots : a_5)$, and let $\tau_5(f)$ be its rational factorization type; for example, $1^3 + 2$ denotes a triple rational factor and an irreducible quadratic factor. Their semilinear orbit counts are $18, 5, 2, 2, 1$; E_f, together in odd characteristic with $\tau_5(f)$, separates the classes up to precisely Frobenius fusion. Table VI prints the canonical representatives, stabilizers, orbit sizes, and fusion lengths.

TABLE VI: Exceptional redundancy-six classes. Each row is one projective-semilinear class. The columns give a canonical syndrome representative, the size and stabilizer order of each constituent $\mathrm{PGL}_2(q)$-orbit, the number ϕ of such orbits fused by coefficient Frobenius, the shared-root pair count E, and the binary-quintic factor type τ_5 in odd characteristic.

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:0:0:1:0:1]$	168	2	1	7	1^3+2
7	$[0:0:1:0:3:1]$	336	1	1	19	1^2+1+2
7	$[0:0:1:0:3:2]$	336	1	1	15	1^2+3
7	$[0:1:0:0:0:1]$	168	2	1	10	$1+2+2$
7	$[0:1:0:0:2:0]$	112	3	1	33	$1+1+3$
7	$[0:1:0:1:1:2]$	336	1	1	9	$1+4$
7	$[0:1:0:1:1:5]$	336	1	1	9	$1+1+3$

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:1:0:1:3:0]$	336	1	1	21	$1+1+1+2$
7	$[0:1:0:1:3:3]$	336	1	1	7	$1+1+3$
7	$[0:1:0:1:3:6]$	336	1	1	14	$1+1+3$
7	$[0:1:0:3:0:5]$	168	2	1	11	$1+4$
7	$[0:1:0:3:0:6]$	168	2	1	20	$1+4$
7	$[0:1:0:3:2:1]$	336	1	1	13	$1+1+3$
7	$[1:0:0:0:1:2]$	336	1	1	7	$1+4$
7	$[1:0:0:0:1:3]$	336	1	1	23	5
7	$[1:0:0:3:1:3]$	336	1	1	13	$1+4$
7	$[1:0:1:0:5:2]$	336	1	1	8	$1+4$
7	$[1:0:1:0:6:2]$	336	1	1	8	5
8	$[0:0:0:1:0:0]$	9	56	1	0	–
8	$[0:1:1:0:2:1]$	504	1	3	20	–
8	$[0:1:1:0:2:5]$	504	1	3	19	–
8	$[1:0:0:1:3:6]$	504	1	3	14	–
8	$[1:0:1:3:5:2]$	168	3	1	54	–
9	$[0:1:0:0:0:4]$	180	4	2	28	$1+4$
9	$[0:1:0:1:4:2]$	720	1	2	29	$1+4$
11	$[0:0:0:1:0:0]$	132	10	1	60	1^3+1^2
11	$[0:1:0:2:0:10]$	660	2	1	30	$1+4$
13	$[0:1:0:0:0:2]$	546	4	1	60	$1+4$

For $q=8$, integers are binary polynomial-basis codes with $\alpha^3=\alpha+1$; for $q=9$, they are ternary polynomial-basis codes with $\alpha^2=-1$. Thus $\sum\phi=(18,11,4,2,1)$ and $\sum\phi|\mathcal{O}|=(5152,4713,1800,792,546)$, in the field order $7,8,9,11,13$.

B. Redundancy seven

For

$$f=(a_0:\ldots:a_6)\in\mathbb{P}(\Gamma^6E),\qquad W_f=\ker\begin{pmatrix}a_0&a_1&a_2&a_3&a_4&a_5\\a_1&a_2&a_3&a_4&a_5&a_6\end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r)=(a_1-ra_0,\ldots,a_6-ra_5).$$

Coefficient collection gives

$$(t-r)g\in W_f\quad\Longleftrightarrow\quad g\in W_{b(r)};$$

the lift is squarefree exactly when g is squarefree and $g(r)\neq0$.

On the rank-two Hankel-row open, the contracted net $W_{b(r)}$ has quadratic gcd exactly when its three-row catalecticant has rank at most two. The rank-one boundary is part of the persistent contained case. Unless the first-polar line is contained in that rank-two locus, the remaining parameters form the degree-at-most-three intersection of Proposition B.2; the contained case is Corollary B.13.

Proposition B.10 (Complete two-marker lower package)**.** Let W_h be a trivial-gcd quartic net and let x be a prescribed forbidden root. Assume in characteristic two that $h\notin\mathcal{N}_3$. For every prime power $q\geq37$, the net contains a split squarefree quartic avoiding x.

Proof. Choosing the second marker. By equivariance choose coordinates in which x is finite; all parameter equations below are then homogenized in s, so the infinity chart is included. Choose the second contraction marker s. The unavailable values of s comprise: $s=x$, of degree one; at most three intersections with the lower secant cubic, by Proposition B.2; at most four intersections with the cyclic surface or its characteristic-three wild replacement; and at most six points of the collision divisor, by Propositions B.4 and B.5. In characteristic two the cyclic replacement is a plane, so its pullback has degree at most one unless the polar line is contained in it, the excluded case $h\in\mathcal{N}_3$, by Proposition B.5. A lower pencil with any fixed factor already lies in the degree-three persistent intersection, by Lemma IV.1. Hence the complete second-step parameter scheme has degree at most

$$1+3+4+6=14<q+1.$$

Choose s outside it.

The S_3 branch. The bottom syndrome lies outside the persistent carrier, so Lemma IV.1 makes its pencil base-point-free; outside the cyclic locus its cover has geometric monodromy S_3. Its off-diagonal curve is geometrically integral of arithmetic genus one. Proposition VII.6 and Lemma VII.7 show that at most twelve rational points encode nonsplit members. Base-point-freeness gives at most one pencil member through each of the distinct markers x, s, and these contribute at most twelve further points. Thus at most 24 rational points of the bottom off-diagonal curve fail to encode a split member avoiding both markers. Aubry–Perret supplies an allowed member when

$$q + 1 - 2\sqrt{q} > 24,$$

whose first prime-power solution is 37. The cubic lifts through s to a split squarefree quartic in W_h avoiding x. □

Proposition B.11 (Collision divisor)**.** After fixed factors are removed, the collision divisor of the moving g^3_5 has degree at most eight.

Proof. This is Lemma A.5 with $j = 7$: the series is separable and $2j - 6 = 8$. □

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is split-free for odd m, and hence deep whenever the separate radius-six gate holds. For even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition B.12 (Central binary lift)**.** The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition B.6. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. □

Corollary B.13 (Exhaustive R7 contained-component classification)**.** The one-step contained-component assertion from redundancy seven to redundancy six is exhaustive in every characteristic. A noninjective contraction is the shallow rank-one boundary. Containment in the lower persistent locus gives upper catalecticant rank at most two; after the shallow rational secants are removed, these are exactly the persistent tangent and conjugate-secant families. The complete lift of the only additional lower component, the characteristic-two nucleus line, is the central point e_3. The remaining collision divisor is nonzero. Thus the nonshallow contained list is precisely the persistent locus and, in characteristic two, e_3; there is no further contained component.

Proof. Apply Proposition A.7 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition B.12. Proposition B.11 shows that the collision divisor is nonzero. These are respectively the contraction-indeterminacy, lower-component-containment, and identical-collision alternatives of Definition A.3, so the list is exhaustive. □

Theorem B.14 (Redundancy-seven classification)**.** Certificate R7 classifies the split-free syndromes for

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}.$$

For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent locus of size $q(q+1)^2/2$, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles, after removing the persistent locus and also the central nucleus singleton at $q = 8$, are: An exponent records multiplicity, so, for example, 84^5 means five distinct orbits of size 84.

7	$56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141}$
8	$63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86}$
9	$180^3,\ 240^6,\ 360^{18},\ 720^{27}$
11	$264^2,\ 440,\ 660^2.$

This is the complete split-free classification for all prime powers $q \geq 7$. Together with the radius and exact-distance statements below, it yields the complete code deep-hole classification. The covering radius is six for $q = 7, 9$ and for every $q \geq 11$. At $q = 8$, where [4, Thm. 17(1)] gives radius seven, the diagonal tangent orbit (nine directions) and the fixed central nucleus are exactly the distance-seven rows; exhaustive replay puts all 45 other frozen representatives and the remaining three persistent representatives at distance six. Thus the $q = 8$ classification is complete. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For fixed irreducible quadratic Q, the persistent fiber has $q+1$ points, while for $Q=\lambda^2$ it has q non-rank-one points. There are $q(q-1)/2$ irreducible quadratics and $q+1$ rational squares. Hence the persistent count is

$$\frac{q(q-1)}{2}(q+1)+(q+1)q=\frac{q(q+1)^2}{2}.$$

For $q\geq 37$, choose a first marker outside the at most three catalecticant intersections, eight collision parameters, and, in characteristic two, one intersection with the nucleus line. The first-marker exclusions put the contracted quartic net outside its persistent intersection; Lemma IV.1 therefore gives trivial gcd. Proposition B.10 supplies the complete second one-step package: it excludes the persistent and cyclic/wild carriers before applying the pointed S_3 cover. These first-marker bounds and Proposition B.10 are exactly the depth-two data of Theorem A.6; the terminal genus-one curve has at most 24 bad rational points. Corollary B.13 leaves exactly the persistent locus and Proposition B.12. Certificate R7 covers

$$q\in\{7,8,9,11,13,16,17,19,23,25,27,29,31,32\}$$

by the exact primary quotient enumeration. A separate replay checks the recorded representatives and their orbits. A direct-locus replay, independent of the R7 generator and quotient partition, imports neither: below 16 it constructs the literal four-secant complement, from 16 onward it constructs the classified R6 pointed locus directly, including the transient marked orbit at $q=19$, and in every field it intersects all contraction conditions and rebuilds the complete $\mathrm{PGL}_2(q)$-orbit and Frobenius partition. It agrees exactly with the public record in all fourteen fields.

For $q\geq 11$, Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-6))=6$. It remains to close $q=7,9$. Here $\mathrm{PRS}(q-6)$ has parameters $[q+1,q-6,8]_q$. If a syndrome f had weight seven, adjoining f to its parity-check matrix would, by (12), define an $[q+2,q-5,8]_q$ MDS code. At $q=7$ this would be an $[9,2,8]_7$ MDS code, whose generator columns would give nine distinct points of $\mathbb{P}^1(\mathbb{F}_7)$, although that line has only eight points. At $q=9$ it would be an $[11,4,8]_9$ MDS code, excluded by Ball–De Beule [23, Thm. 5.2]: here $9=3^2$ and $4=2\cdot 3-2$, the endpoint of their theorem. Thus every syndrome has weight at most six in both fields. The nonempty split-free lists above give the reverse inequality, so the radius is six. Together with the cited $q=8$ radius and exact-distance replay, this promotes the split-free classification over every prime power $q\geq 7$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records. There are respectively $194,119,54,5$ exceptional representatives at $q=7,8,9,11$; the complete lists are printed in `supplement/CLASSIFICATION-RECORDS.md`, while the matching JSON is the machine-readable record. The following gives one printed canonical representative for every orbit-size block in the displayed profile; repeated orbits within a block are distinguished in the full list:

q	$\lvert\mathcal{O}\rvert$	representative
7	56	$[0:0:0:0:1:0:0]$
	84	$[0:1:0:0:0:1:0]$
	112	$[0:0:1:0:0:2:0]$
	168	$[0:0:0:0:1:0:1]$
	336	$[0:0:0:1:0:1:1]$
8	63	$[0:0:0:1:0:0:1]$
	72	$[0:0:0:0:1:0:0]$
	84	$[0:1:1:1:1:3:6]$
	168	$[1:0:1:1:3:7:7]$
	252	$[0:0:1:0:1:0:0]$
	504	$[0:0:1:0:1:0:1]$
9	180	$[0:1:0:0:0:4:0]$
	240	$[0:0:1:0:1:1:1]$
	360	$[0:1:0:1:3:3:2]$
	720	$[0:0:1:1:0:2:2]$
11	264	$[0:1:0:0:0:0:2]$
	440	$[1:0:1:1:4:3:3]$
	660	$[1:0:1:0:1:4:8]$.

For $q=8,9$, the coordinates use the same polynomial-basis encoding specified below Table VI.

The $q=19$ lower net

$$W=\langle 1,t^3,t^4\rangle$$

shows why the parameter must remain marked. Its split squarefree members are exactly

$$U(T^3-uU^3),\qquad u\in(\mathbb{F}_{19}^{\times})^3=\{1,7,8,11,12,18\}.$$

All six contain the marked point at infinity. The obstruction is pointed rather than intrinsic to W: at the marker 0, for example, $U(T^3 - U^3)$ is an admissible member. No coherent sextic polar line lies in the infinity-pointed orbit. Thus the R7 proof escapes over the marker parameter; classifying unpointed bad lower fibers would retain a false exception.

TABLE VII
FIELD-RANGE CLOSURE LEDGER. "CERTIFICATE" MEANS A COMPLETE FINITE CLASSIFICATION WITH CANONICAL REPRESENTATIVES, NOT MERELY AN ORBIT COUNT.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 17, 19$	direct census and independent replay
	16	characteristic-two branch budget, Corollary VII.9; certificate row retained as a regression check
	$q \geq 23$	cubic-cover geometry, Lemma VII.10
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. The covering-radius argument in Theorem B.14 promotes them to code deep-hole directions over every listed field. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

There is no prime power in any of the open numerical intervals $(19, 23)$, $(27, 29)$, or $(32, 37)$. Thus adjacent finite and geometric rows in Table VII leave no unlisted field order.

APPENDIX C
VERIFICATION AND FORMAL BOUNDARY

No geometric classification theorem in this paper is claimed to be formalized in Lean. The formal layer checks coordinate identities and conditional synthesis implications with the component geometry, rational-point existence, group actions, and certificate semantics kept as explicit inputs.

The bounded finite classifications above are accompanied by machine-checkable certificates. Table VIII records exactly which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Section II summarizes the corresponding mathematical mechanisms and closing gates.

TABLE VIII: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen audited field records; seven-field finite bridge, with $q = 16$ covered as a regression check rather than a dependency	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	primary quotient enumeration and representative/orbit replay; split-free and radius flags remain separate
Certificate R7 direct locus	all fourteen R7 certificate fields	candidate-domain and marker-intersection reconstruction, orbit and Frobenius replay; shares the direct-locus engine, R5 field layer, and R6 pointed theorem from $q \geq 16$, and is not a second field implementation
Certificate SC	terminal factorization and coherent-Fano identities	Python and Singular elimination replays; density and finite-component selection are separately kernel checked

Table VI is a deterministic projection of the hash-pinned public R6 and R6-NF records. The top-level verifier regenerates it and checks byte-for-byte equality with the included source.

The supplement's `CERTIFICATE-SCHEMA.md` defines the public labels and replay classes; its reproduction guide and two manifests record commands, locks, hashes, and byte counts. The entry point `supplement/verify.py` checks those records, the classification projections, and the local PDF; `--replay` runs the paper-local Python replays.

The exact Lean declarations, hypotheses, axiom audit, and replay command are listed in `LEAN-STATEMENTS.md`. They check the coordinate, budget, density, closure, component-selection, and conditional-synthesis interfaces. Concrete component geometry, fixed-level carrier arithmetic, classical radius inputs, and certificate semantics remain explicit inputs. The formal layer also checks the field-eight record but draws no quantum consequence from it, because the required one-column MDS extension does not exist at radius $r-1$ (Remark VII.15).

A. Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. This appendix assigns each computational or formal claim its exact trust route. The supplement is part of the submission package. The accepted artifact will additionally record its release commit, archive hash, and version-specific DOI in `supplement/RELEASE-MANIFEST.md`; the stable concept DOI is 10.5281/zenodo.21682069.

The electronic supplement includes Projective Reed–Solomon Toolkit, a proof-carrying executable for exact canonicalization, budgeted distance and decoding, theorem-gated classification, certificate replay, and the simultaneous locator construction of Theorem V.4. It fails closed on budget or theorem-domain exhaustion: computation outside a proved classification domain cannot create a positive deep-hole claim. Its README specifies the commands, complexity, registry, and trust semantics.

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